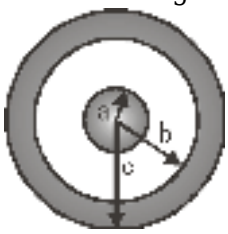
PHYSICS

1) A solid conducting sphere of radius a has a net positive charge 2Q. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge –Q. The surface charge density on the inner and outer surfaces of the spherical shell will be :-



(1) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(2) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(3) $0, \frac{Q}{4\pi c^2}$

(4) None of the above

2) Two small conducting spheres of equal radius have charges +10μC and –20μC respectively & when placed at a distance R from each other experience force $\vec{F}_1$. If they are brought in contact and separated to the same distance, they experience force $\vec{F}_2$. The ratio of $|\vec{F}_1|$ to $|\vec{F}_2|$ is :-

(1) 1 : 8

(2) 8 : 1

(3) 1 : 2

(4) 2 : 1

3) If initially flux through hemisphere is zero. Now q_1 and q_2 placed according to figure. Due to which charge flux through hemisphere will change :-



(1) Only q_1

(2) Only q_2

(3) Both

(4) None

4) Two charges of 5Q and –2Q are situated at the points (3a, 0) and (–5a, 0) respectively. The electric flux through a sphere of radius '4a' having center at origin is :

(1) $\frac{2Q}{\varepsilon_0}$

(2) $\frac{5Q}{\varepsilon_0}$

(3) $\frac{7Q}{\varepsilon_0}$

(4) $\frac{3Q}{\varepsilon_0}$

5) The electrostatic potential on the surface of a charged conducting sphere is 100V. Two statements are made in this regard:
S_1 : At any point inside the sphere, electric intensity is zero.
S_2 : At any point inside the sphere, the electrostatic potential is 100V.
Which of the following is a correct statement ?

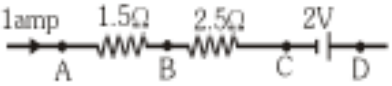
(1) S_1 is true but S_2 is false.

(2) Both S_1 & S_2 are false.

(3) S_1 is true, S_2 is also true and S_1 is the cause of S_2.

(4) S_1 is true, S_2 is also true but the statements are independent.

6) In the circuit element given here, if the potential at point B, $V_B = 0$, then the potentials of A and D are given as:-



(1) $V_A = -1.5V$, $V_D = +2V$

(2) $V_A = +1.5V$, $V_D = +2V$

(3) $V_A = +1.5V$, $V_D = +0.5V$

(4) $V_A = +1.5V$, $V_D = -0.5V$

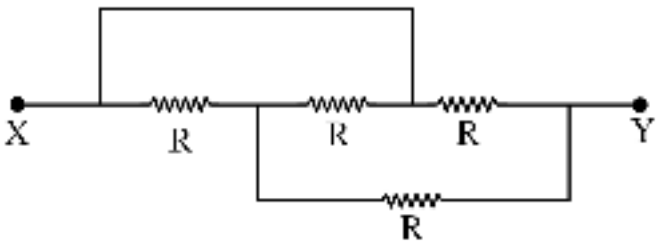
7) Three resistance P, Q, R each of 2Ω and an unknown resistance S form the four arms of a Wheatstone bridge circuit. When a resistance of 6Ω is connected in parallel to S the bridge gets balanced. What is the value of S ?

(1) 2 Ω

(2) 3 Ω

(3) 6 Ω

(4) 1 Ω



8) Calculate equivalent resistance across X & Y.

(1) $\frac{5R}{3}$

(2) $\frac{3R}{5}$

(3) $\frac{6R}{5}$

(4) $\frac{7R}{5}$

9) A moving coil galvanometer has a resistance of 50 Ω and gives full scale deflection for 10mA. How could it be converted into an ammeter with a full scale deflection for 1A.

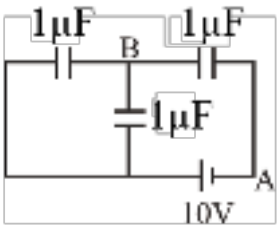
(1) 50/99 Ω should be connected in series

(2) 50/99 Ω should be connected in parallel

(3) 0.01 Ω should be connected in series

(4) 0.01 Ω should be connected in parallel

10) If potential of A is 10V, then potential of B is -



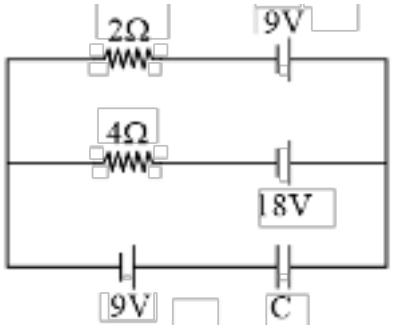
(1) 25/3 V

(2) 50/3 V

(3) 100/3 V

(4) 50 V

11) In the circuit shown, calculate the potential drop (in volt) across the capacitor.



(1) 1V

(2) 2V

(3) 3V

(4) 4V

12) A long straight wire along the z-axis carries a current I in the negative z direction. The magnetic field $\vec{B}$ at a point having coordinates (x, y) in the z = 0 plane is :-

(1) $\frac{\mu_0 I(y\hat{i}-x\hat{j})}{2\pi(x^2+y^2)}$

(2) $\frac{\mu_0 I(x\hat{i}+y\hat{j})}{2\pi(x^2+y^2)}$

(3) $\frac{\mu_0 I(x\hat{j}-y\hat{i})}{2\pi(x^2+y^2)}$

(4) $\frac{\mu_0 I(x\hat{i}-y\hat{j})}{2\pi(x^2+y^2)}$

13) An electron revolving with period 'T' in orbit of radius 'R'. This electron is equivalent to a magnetic dipole, then dipole moment of this electron is :-

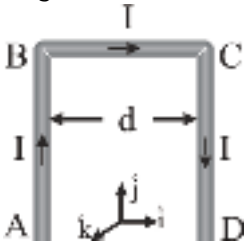
(1) $\frac{\pi Re}{T}$

(2) pR^2eT

(3) $\frac{\pi R^2 e}{T}$

(4) $\frac{2\pi Re}{T}$

14) AB and CD are long straight conductor, distance d apart, carrying a current I. The magnetic field at the midpoint of BC is :-



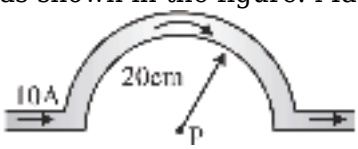
(1) $\frac{-\mu_0 I}{2\pi d}\hat{k}$

(2) $\frac{-\mu_0 I}{\pi d}\hat{k}$

(3) $\frac{-\mu_0 I}{4\pi d}\hat{k}$

(4) $\frac{-\mu_0 I}{8\pi d}\hat{k}$

15) A current of 10A is passing through a long wire which has semicircular loop of the radius 20cm as shown in the figure. Magnetic field produced at the centre of the loop is :



(1) 10πμT

(2) 5πμT

(3) 4πμT

(4) 2πμT

16) A charged particle moves through a magnetic field perpendicular to its direction.Then-

(1) the momentum changes but the kinetic energy is constant

(2) both momentum and kinetic energy of the particle are not constant

(3) both momentum and kinetic energy of the particle are constant

(4) kinetic energy changes but the momentum is constant

17) An electron moving in a circular orbit of radius 'r' makes 'υ' revolutions per second. The magnetic field produced at the centre has a magnitude of :

(1) $\frac{\mu_0 \upsilon e}{2r}$

(2) $\frac{\mu_0 \upsilon^2 e}{2r}$

(3) $\frac{\mu_0 \upsilon e}{2\pi r}$

(4) Zero

18) Two coils have mutual inductance 0.005 H. the current changes in the first coil according to equation $I = I_0 \sin \omega t$, where $I_0 = 10A$ and ω = 100π rad/s. the maximum value of EMF in the second coil is-

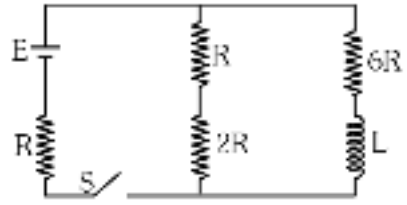
(1) 2π

(2) 5π

(3) π

(4) 4π

19) If switch is closed at t = 0, the current supplied by battery immediately after closing the switch is



(1) $\frac{E}{3R}$

(2) $\frac{E}{4R}$

(3) $\frac{E}{7R}$

(4) $\frac{E}{R}$

20) The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field 10^{-3} Wb/m^2 acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μV, is :-

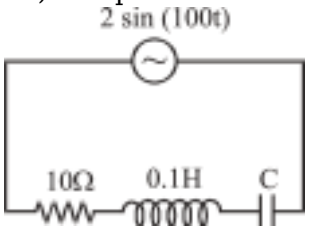
(1) $\frac{2}{\pi}$cm

(2) $\frac{3}{\pi}$cm

(3) $\frac{4}{\pi}$cm

(4) $\frac{5}{\pi}$cm

21) The power factor of circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to :-



(1) 400 µF

(2) 300 µF

(3) 500 µF

(4) 200 µF

22) An AC voltage is represented by :-

E = $575\sqrt{5}\cos(75\pi t)$

How many times will the current become zero in 1 second ?

(1) 75 times

(2) 50 times

(3) 100 times

(4) 575 times

23) The electric field part of an electromagnetic wave in vacuum is

$$E = 3.1\frac{N}{C}\cos\left[\left(1.8\frac{rad}{m}\right)y + \left(5.4\times10^{8}\frac{rad}{s}\right)t\right]\hat{i}$$

The wavelength of this part of electromagnetic wave is

(1) 1.5 m

(2) 2 m

(3) 2.5 m

(4) 3.5 m

24) Match column I with column II and select the correct option from the choices given below the columns.

Column I		Column II	
(A)	$\oint_S \vec{E}.\vec{ds} = \frac{q}{\varepsilon_0}$	(P)	Faraday's law of EMI
(B)	$\oint_S \vec{B}.\vec{ds} = 0$	(Q)	Gauss's law in magnetism

(C)	$\oint_s \vec{E}.d\vec{\ell} = -\frac{d\phi_B}{dt}$	(R)	Maxwell-Ampere's circuital law
(D)	$\oint \vec{B}.d\vec{\ell} = \mu_0\left[I_c + \varepsilon_0\frac{d\phi}{dt}\right]$	(S)	Gauss's law in electrostatics

(1) (A)→(Q); (B)→(S); (C)→(R); (D)→(P)

(2) (A)→(P); (B)→(S); (C)→(Q); (D)→(R)

(3) (A)→(S); (B)→(Q); (C)→(P); (D)→(R)

(4) (A)→(R); (B)→(Q); (C)→(S); (D)→(P)

25) An equilateral prism deviates a ray through 40° for two angles of incidence differing by 20°. The possible angle of incidence is :

(1) 40°

(2) 50°

(3) 20°

(4) 70°

26) **Assertion :** The minimum distance between a real object and its real image formed by a concave mirror is 4f.
Reason : Convex mirror form a virtual and inverted image for all positions of the real object.

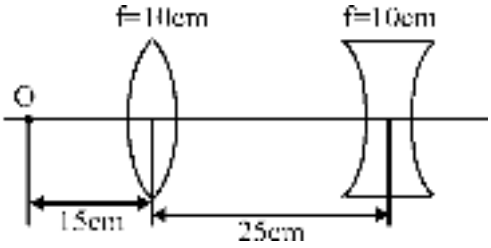
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

27) Find the final image from concave lens.



(1) 10 cm left

(2) 10 cm right

(3) 5 cm left

(4) 5 cm right

28) A ray is travelling parallel to principal axis of a concave mirror of focal length f. If the distance between ray and principal axis is h, magnitude of deviation of incident ray produced is (h<<f) :-

(1) $\pi - \frac{h}{f}$

(2) $\pi - \frac{2h}{f}$

(3) $\frac{h}{f}$

(4) $\frac{2h}{f}$

29) A container contains a liquid with refractive index of 1.2 up to a height of 60 cm and another liquid having refractive index 1.6 is added to height H above first liquid. If viewed from above, the apparent shift in the position of bottom of container is 40 cm. The value of H is ___ cm. (Consider liquids are immisible) :

(1) 60

(2) 80

(3) 90

(4) 100

30) A light ray enters a solid glass sphere of refractive index $\mu = \sqrt{3}$ at an angle of incidence 60°. The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degrees) between the reflected and refracted rays at this surface is_____.

(1) 30

(2) 45

(3) 60

(4) 90

31) In Young's double slit experiment, fringe order is represented by m, then fringe width is :-

(1) Independent of m

(2) Directly proportional to m

(3) Directly proportional to (2m + 1)

(4) Inversely proportional to (2m + 1)

32) In YDSE, two slits are separated by 3mm and illuminated by light 480 nm. The screen is at a distance of 2m from the plane of slit. Calculate the separation between 3rd bright fringe and the 1st dark fringe:-

(1) 0.8 mm

(2) 0.4 mm

(3) 0.6 mm

(4) 0.5 mm

33) Consider an interference pattern between two coherent sources. If I_1 and I_2 be intensities at points where the phase difference are $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ respectively, then the intensity at maxima is

(1) $\frac{I_2 - 3I_1}{2}$

(2) $\frac{I_1 - 3I_2}{2}$

(3) $\frac{3I_1 - I_2}{2}$

(4) $\frac{3I_1 - I_2}{4}$

34) Angular width of the central maxima in the Fraunhofer diffraction for λ = 6000 Å is θ_0. When the same slit is illuminated by another monochromatic light, the angular width decreases by 70%. The wavelength of this light is,

(1) 1800 Å

(2) 4200 Å

(3) 6000 Å

(4) 420 Å

35) The binding energy per nucleon of Cu^{63} is 8.75 MeV and of Cu^{64} is 8.73 MeV. The energy (in MeV) required to remove a neutron from Cu^{64} is :-

(1) 12.6

(2) 3.64

(3) 4.23

(4) 7.47

36) The shortest wavelength in the Lyman series of hydrogen spectrum is 912 Å corresponding to a photon energy of 13.6 eV. The shortest wavelength in the Balmer series is:

(1) 3648 Å

(2) 228 Å

(3) 6566 Å

(4) 8208 Å

37) There are n_1 photons of frequency ν_1 in a beam of light. In an equally energetic beam, there are n_2 photons of frequency ν_2 . Then the correct relation is

(1) $\frac{n_1}{n_2} = 1$

(2) $\frac{n_1}{n_2} = \frac{\nu_1}{\nu_2}$

(3) $\frac{n_1}{n_2} = \frac{\nu_2}{\nu_1}$

(4) $\frac{n_1}{n_2} = \frac{\nu_1^2}{\nu_2^2}$

38) A photoelectric effect supports quantum nature of light because :
(A) there is a minimum frequency of light below which no photoelectrons are emitted.
(B) the maximum kinetic energy of photo electrons depends only on the frequency of light and not on its intensity
(C) Above threshold frequency even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately

(D) electric charge of photoelectrons is quantized In the above true statement is :-

(1) A, B, C
(2) A, B
(3) A
(4) All

39) The de-Broglie wavelength of a particle accelerated with 150 volt potential is 10^{-10} m. If it is accelerated by 600 volts potential difference, its wavelength will be :-

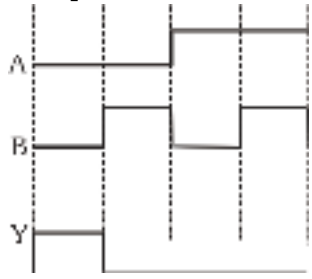
(1) 0.25 Å
(2) 0.5 Å
(3) 1.5 Å
(4) 2 Å

40) In figure V_0 is the potential barrier across a p-n junction, when no battery is connected across the junction



(1) 1 and 3 both correspond to forward bias of junction
(2) 3 corresponds to forward bias of junction and 1 corresponds to reverse bias of junction
(3) 1 corresponds to forward bias and 3 corresponds to reverse bias of junction.
(4) 3 and 1 both correspond to reverse bias of junction.

41) There is a gate with two inputs A and B, and one output Y. Using the input and output wave forms, identify the gate :-



(1) NAND
(2) NOR
(3) OR
(4) AND

42) **Statement - I :-** Photodiode is used in reverse bias.
Statement-II:- In reverse bias change in current is due to additional minority charge carriers. Here due to photoeffects fractional change in reverse current is easily measurable.

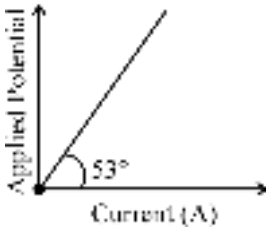
(1) Statement - I and II both are correct.

(2) Statement - I and II both are incorrect.

(3) Statement - I is correct and statement - II is incorrect.

(4) Statement - I is incorrect and statement - II is correct.

43) The variation of applied potential and current flowing through a given wire is as shown in figure. The length of the wire is 50 cm and diameter of wire is measured to be 0.2 cm. The resistivity of the wire, is given near to :



(1) 2×10^{-4} Ω-m

(2) 4×10^{-6} Ω-m

(3) 8×10^{-6} Ω-m

(4) 4×10^{-4} Ω-m

44) Match the terms given in list-I with suitable option provided in list-II, regarding experiment to calculate resistance and figure of merit of a galvanometer using half deflection method :

	List-I		**List-II**
A.	Figure of merit (k)	p	In series
B	Current sensitivity (SI)	Q	To get power loss minimized
C	Deflection is made half by using resistance	R	In shunt/parallel
D	High resistance Box	S	Current per unit deflection
		T	Deflection per unit current

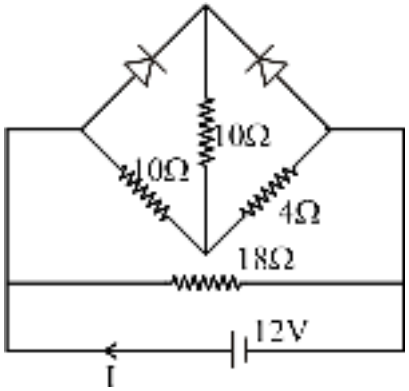
(1) A - S, B - T, C - R, D - P

(2) A - T, B - S, C - R, D - P

(3) A - T, B - S, C - P, D - R

(4) A - S, B - T, C - P, D - Q

45) In circuit containing ideal diodes, the current through battery will be :



(1) 0.5 A

(2) 1.2 A

(3) 2 A

(4) 2.4 A

CHEMISTRY

1) Reaction A → B, r = k[A]
occurs in two different conditions with and without catalyst then select the correct statement

(1) Pre–exponetial factor in relation $k = Ae^{-Ea/RT}$ will be different in both cases.

(2) On increasing temperature rate in absence of catalyst will increase more rapidly as compared to in presence of catalyst.

(3) Activated complex that will form in both case will be same.

(4) Reaction should be an elementary reaction.

2) For reaction A + B → C

Step -I $A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} I$

Step -II $I \xrightarrow{k_3} C$

if value of k_1, k_2 & k_3 are $2 \times 10^3 \frac{litre}{mol}.sec^{-1}$, 10^3 sec^{-1} & 0.01 sec^{-1} then value of overall rate constant for reaction?

(1) $0.01 \frac{mol}{litre} sec^{-1}$

(2) 0.01 $sec.^{-1}$

(3) $0.02 \frac{litre}{mol}.sec^{-1}$

(4) $0.02 \frac{mol}{litre} sec^{-1}$

3) For the reaction $CaCO_3 \rightarrow CaO + CO_2$ if rate of disappereance of $CaCO_3$ is 240 g s^{-1} then find rate of formation of CO_2 .

(1) 105.6 g s^{-1}

(2) 240.6 g s^{-1}

(3) 120.6 g s^{-1}

(4) 95.6 g s^{-1}

4)

A(g) → 2B (g) + C (g) (first order reaction)

Time (in min)	0	100
Total pressure (in mmHg)	100	200

The half life period of reaction is -

(1) 50 min

(2) 100 min

(3) 70 min

(4) Can not predict

5) A galvanic cell is composed of two hydrogen electrodes, one of which is a standard. In which of the following solutions should the other electrode be immersed to get maximum emf ?
K_a (CH_3COOH) = 2×10^{-5} , K_a (H_3PO_4) = 10^{-3} .

(1) 0.1 M HCl

(2) 0.1 M CH_3COOH

(3) 0.1 M H_3PO_4

(4) 0.1 M H_2SO_4

6) The standard reduction potentials of some half cell reactions are given below :

$PbO_2 + 4H^+ + 2e^- \rightleftharpoons Pb^{2+} + 2H_2O$	E^0 = 1.455 V
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	E^0 = 1.51 V
$Ce^{4+} + e^- \rightleftharpoons Ce^{3+}$	E^0 = 1.61 V
$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$	E^0 = 1.71 V

Pick out the incorrect statement :

(1) Ce^{+4} will oxidise Pb^{2+} to PbO_2

(2) MnO_4^- will oxidise Pb^{2+} to PbO_2

(3) H_2O_2 will oxidise Mn^{+2} to MnO_4^-

(4) PbO_2 will oxidise Mn^{+2} to MnO_4^-

7) The conductivity of 0.2M solution of KCl at 298 K is 0.0248 S cm^{-1}. Calculate its molar conductivity.

(1) 248 S cm^2 mol^{-1}

(2) 124 S cm^2 mol^{-1}

(3) 12.4 S cm^2 mol^{-1}

(4) 24.8 S cm^2 mol^{-1}

8) A hydrogen electrode is immersed in a solution with pH = 0 (HCl). By how much will the potential (reduction) change if an equivalent amount of NaOH is added to the solution. (Take p_{H_2} = 1 atm) T = 298 K.

(1) increase by 0.41V

(2) increase by 59 mV

(3) decrease by 0.41V

(4) decrease by 59 mV

9) What will be molar conductivity of Al^{+3} ions at infinite dilution if molar conductivity of $Al_2(SO_4)_3$ and SO_4^{-2} is 858, 160 $S \cdot cm^2 mol^{-1}$ respectively at infinite dilution ?

(1) 189 $S \cdot cm^2 mol^{-1}$
(2) 698 $S \cdot cm^2 mol^{-1}$
(3) 1018 $S \cdot cm^2 mol^{-1}$
(4) 429 $S \cdot cm^2 mol^{-1}$

10) A sample of air is saturated with benzene (vapour pressure = 100 mm Hg at 298 K) at 298K, 750mm Hg pressure. If it is isothermally compressed to one third of its initial volume, the final pressure of the system is

(1) 2250 torr
(2) 2150 torr
(3) 2050 torr
(4) 1950 torr

11) Two liquids A and B have vapour pressure of 0.6 bar and 0.2 bar in pure state respectively. In an ideal solution of these two calculate mole fraction of A at which these two liquid have equal partial vapour pressure :–

(1) 0.75
(2) 0.25
(3) 0.33
(4) 0.67

12) The cryoscopic constant for acetic acid is 3.6 K kg/mol. A solution of 1 g of a hydrocarbon in 100 g of acetic acid freezes at 16.14°C (if freezing point of acetic acid is 16.60°C). The hydrocarbon contains 92.3% carbon. The molecular formula of hydrocarbon is C_XH_Y then X + Y is :

(1) 2
(2) 12
(3) 5
(4) 8

13) **Assertion :** On adding NaCl to water, the boiling point of water increases.
Reason : On adding NaCl to water, the vapour pressure of water increases.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.
(2) Assertion is correct but reason is incorrect.
(3) Both assertion and reason are correct but reason is not the correct explanation of assertion.
(4) Assertion is incorrect but Reason is correct

14) Which one of following statements is incorrect in respect of lyophilic sols ?

(1) There is a considerable interaction between the dispersed phase and dispersion medium

(2) These are quite stable and are not easily coagulated

(3) They are negatively charged sols.

(4) The particle are hydrated

15)

The following facts are available :
$2X^- + Y_2 \rightarrow 2Y^- + X_2$
$2W^- + Y_2 \rightarrow$ No reaction
$2Z^- + X_2 \rightarrow 2X^- + Z_2$
Which of the following statement is correct ?

(1) $E^o_{W^-/W_2} > E^o_{Y^-/Y_2} > E^o_{X^-/X_2} > E^o_{Z^-/Z_2}$

(2) $E^o_{W^-/W_2} < E^o_{Y^-/Y_2} < E^o_{X^-/X_2} < E^o_{Z^-/Z_2}$

(3) $E^o_{W^-/W_2} < E^o_{Y^-/Y_2} > E^o_{X^-/X_2} > E^o_{Z^-/Z_2}$

(4) $E^o_{W^-/W_2} > E^o_{Y^-/Y_2} < E^o_{X^-/X_2} < E^o_{Z^-/Z_2}$

16) Which of following is Acetylide ion :

(1) C^{2-}

(2) C_2^-

(3) C_2^{2-}

(4) C_3^{4-}

17)

Choose incorrect match :-

	Compound	**Dimer form through**
(1)	NO_2	2C-2e$^-$ bond
(2)	BH_3	3C-2e$^-$ bond
(3)	$AlCl_3$	2C-2e$^-$ bond
(4)	CH_3COOH	Hydrogen bond

(1) 1

(2) 2

(3) 3

(4) 4

18) Which of following occurs as a consequence of inert pair effect :-
(a) $SnCl_4$ act as a reducing agent
(b) $SnCl_2$ act as an oxidising agent
(c) SnO_2 is amphoteric

(d) CCl_2 is unstable but $PbCl_2$ is stable.
(e) PbO_2 is a good oxidising agent.

(1) a, d, e
(2) d, e
(3) a, b, c, d, e
(4) a, b, c

19) Which of the statements is not true?

(1) $K_2Cr_2O_7$ solution in acidic medium is orange
(2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7
(3) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed
(4) The bond angle of Cr-O-Cr < O-Cr-O in $K_2Cr_2O_7$

20) **Assertion (A)**:- Cr has greater enthalpy of atomisation than that of Zn.
Reason (R) :- Greater the unpaired e^- in d-orbitals favourable for strong interatomic interaction.

(1) Both (A) and (R) are correct and (R) is the correct explanation for (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation for (A)B
(3) (A) is correct but (R) is incorrect
(4) (A) is incorrect but (R) is correct

21) The pair of lanthanoid ions which are diamagnetic is

(1) Ce^{4+} and Yb^{2+}
(2) Ce^{3+} and Eu^{2+}
(3) Gd^{3+} and Eu^{3+}
(4) Pm^{3+} and Sm^{3+}

22) Select correct order :-

(1) Mn > Cr ; Covalent radius
(2) Mn < Cr ; Metallic radius
(3) Cu < Zn ; Density
(4) Cu > Zn ; Melting point

23) On the basis of Sidgwick EAN rule, which of the following statement is not correct ?

(1) $[Mn(CO)_6]$ can act as reducing agent
(2) $[V(CO)_6]$ can act as oxidising agent.
(3) $[Fe(CO)_5]$ can act as both oxidising agent and reducing agent.
(4) $[Mn(CO)_5]$ show dimerisation to gain stability.

24) Which of the following will have two stereoisomeric forms ?
(I) $[Cr(NO_3)_3(NH_3)_3]$ (II) $K_3[Fe(C_2O_4)_3]$

(III) $[CoCl_2(en)_2]^+$ (IV) $[CoBrCl(OX)_2]^{3-}$

(1) I only
(2) I and II
(3) III and IV
(4) All of these

25) Select the correct statement of $3d^4$ to $3d^7$ complex:-

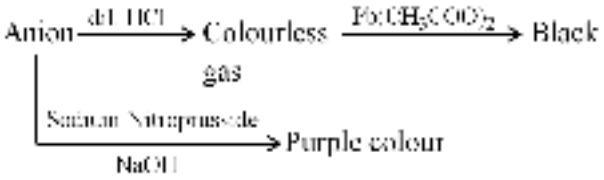
(1) If $\Delta_0 >$ P.E then complex is d^2sp^3 hybridised
(2) If $\Delta_0 <$ P.E then complex is sp^3d^2 hybridised
(3) If $\Delta_0 >$ P.E then ligands are strong field
(4) All of these

26)

Which of the following order is not correct :-

(1) $[Fe(H_2O)_6]^{+3} < [Fe(CN)_6]^{-3}$: Δ_0 order
(2) $[Fe(CN)_6]^{-4} < [Fe(CN)_6]^{-3}$: Fe-C Bond length
(3) $CN^- < CO$: π-accepting tendency
(4) $C_2H_4 >$ $K[PtCl_3(C_2H_4)]$: C-C bond length

27)



Anion is :

(1) CO_3^{-2}
(2) S^{2-}
(3) NO_2^-
(4) NO_3^-

28)

Nessler reagent is :

(1) $NaHgCl_4$
(2) K_2HgI_4
(3) $Hg(NH_3)_2Cl$
(4) K_2HgI_4 + KOH

29) Match the columns-

Column-I (Radical)		Column-II (Effect of Group Reagent)	
(a)	NH_4^+	(P)	Gives yellow ppt with H_2S
(b)	Fe^{+3}	(Q)	Gives white ppt with dil HCl
(c)	Cd^{+2}	(R)	Gives pungent smell gas with NaOH
(d)	Ag^+	(S)	Gives red-brown ppt with NH_4OH

(1) a–R, b–S, c–P, d–Q

(2) a–P, b–Q, c–R, d–S

(3) a–S, b–R, c–Q, d–P

(4) a–R, b–P, c–Q, d–R

30) Match the List-I with List-II -

List-I (Cation)		List-II (Group reagent)	
(A)	Pb^{+2}, Cu^{+2}	(i)	H_2S + HCl
(B)	Al^{+3}, Fe^{+3}	(ii)	$(NH_4)_2CO_3$ + NH_4OH
(C)	Co^{+2}, Ni^{+2}	(iii)	NH_4OH + NH_4Cl
(D)	Ba^{+2}, Ca^{+2}	(iv)	H_2S + NH_4OH

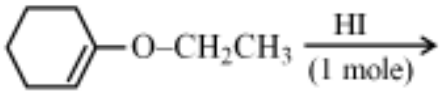
Correct match is :

(1) A–iv; B–ii; C–iii; D–i

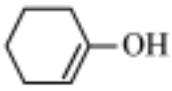
(2) A–i; B–iii; C–iv; D–ii

(3) A–iii; B–i; C–iv; D–ii

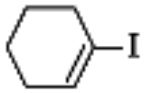
(4) A–i; B–iii; C–ii; D–iv

31) 

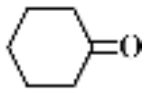
Major products are :

(1) 

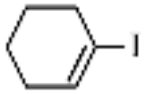
and CH_3CH_2–I

(2) 

and CH_3CH_2–OH

(3) 

and CH_3CH_2–I

(4) 

and CH_3CH_2–I

32) C_2H_5MgCl on reaction with acetone followed by protonation gives produces :

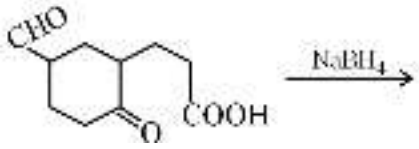
(1) 1° alcohol
(2) 2° alcohol
(3) 3° alcohol
(4) Isobutylene

33) Which of the following compound on stephen reduction can give aldehyde ?

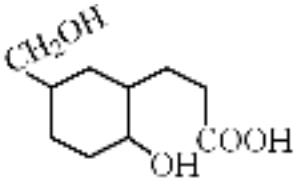
(1) $CH_3CH_2COOC_2H_5$
(2) $CH_3CH_2CONH_2$
(3) CH_3CH_2CN
(4) $CH_3–CH_2–COOH$

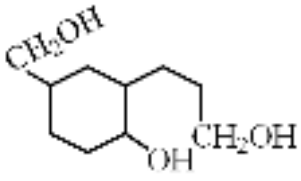
34) $C_2H_5–Cl \xrightarrow[\Delta]{AgCN}$ Major product (X)
Functional isomer of 'X' is :

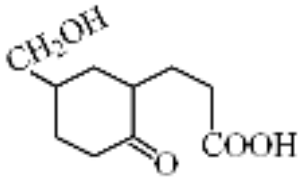
(1) C_2H_5NC
(2) C_2H_5CN
(3) $C_2H_5NO_2$
(4) C_2H_5ONO

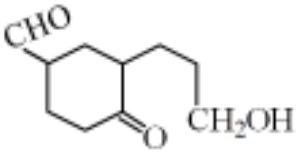
35) 

The product is :

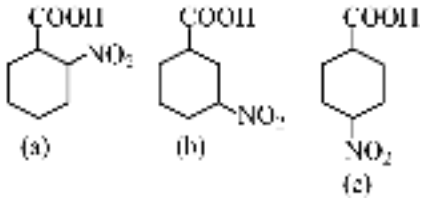
(1) 

(2) 

(3) 

(4) 

36) The correct order of rate of decarboxylation is :



(1) a > b > c

(2) a > c > b

(3) c > b > a

(4) a ≈ b ≈ c

37) Which of the following is soluble in aq. NaOH ?

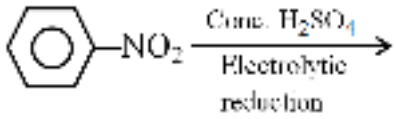
(1) $Ph–SO_2NH–CH_3$

(2) $Ph–SO_2N(CH_3)–CH_3$

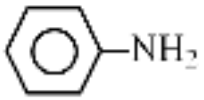
(3) $Ph–SO_2N(C_2H_5)–C_2H_5$

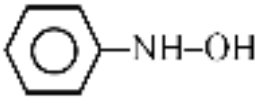
(4) $Ph–SO_2N(CH_3)–C_2H_5$

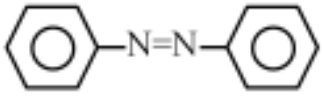
38)

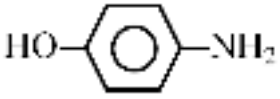


The product is :

(1) 

(2) 

(3) 

(4) 

39) In Kjeldahl's method, NH_3 evolved from 0.50 gm of an organic compound is absorbed in 60 mL $\frac{N}{2}$ H_2SO_4. Unused acid is neutralized by 100 mL $\frac{N}{10}$ NaOH. Calculate % of nitrogen.

(1) 28%

(2) 56%

(3) 14%
(4) 84%

40) Which of the following compound does not give Lassaigne's test for nitrogen ?

(1) C_2H_5–NO_2
(2) C_2H_5–NH_2
(3) NH_2–NH_2
(4) 1 and 2 both

41) Lactose is composed of :

(1) β-D-galactose + β-D-glucose
(2) α-D-galactose + α-D-glucose
(3) α-D-galactose + β-D-glucose
(4) β-D-galactose + α-D-glucose

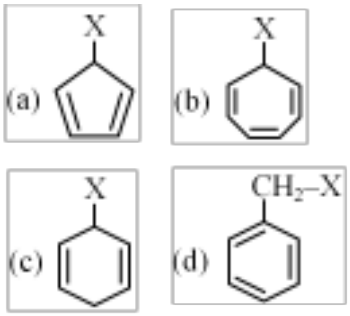
42) Disease 'CHEILOSIS' caused by the deficiency of vitamin :

(1) Thiamine
(2) Riboflavin
(3) Pyridoxime
(4) Ascorbic acid

43) Phenyl acetaldehyde and acetaldehyde cannot be distinguished by :

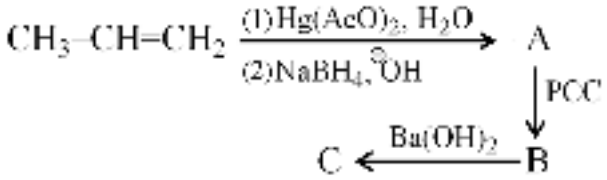
(1) I_2 + NaOH
(2) $NaHSO_3$
(3) Fehling solution
(4) Both (2) and (3)

44) The correct decreasing order of reactivity of S_N1 is :

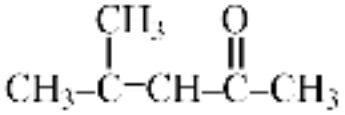


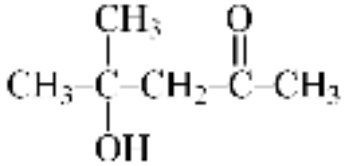
(1) a > b > c > d
(2) c > d > b > a
(3) b >d > c > a
(4) d > b > c > a

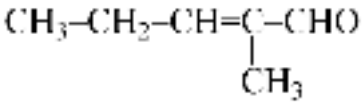
45)

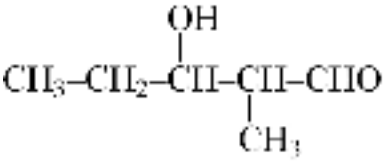


Compound 'C' is :

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1)

Identify the correct match from the column I, II & III.

Column-I		Column-II		Column-III	
1	Dicot stem	a	Polyarch	i	Well developed pith
2	Monocot stem	b	Sunflower	ii	Less developed pith
3	Monocot root	c	Maize	iii	Starch sheath
4	Dicot root	d	2 to 4 xylem patches	iv	Scattered vascular bundle

(1) 1-b-iii, 2-c-iv, 3-a-ii, 4-d-i

(2) 1-b-iii, 2-c-iv, 3-a-i, 4-d-ii

(3) 1-a-iii, 2-c-iv, 3-b-i, 4-d-ii

(4) 1-c-ii, 2-b-i, 3-a-iii, 4-d-iv

2)

Which type of primary xylem is present in given figure ?



(1) Exarch

(2) Endarch

(3) Mesarch

(4) Concentric

3) Match the following :

Column-I		Column-II	
(a)	Hypodermis in dicot stem	(i)	Absent
(b)	Pericycle in dicot stem	(ii)	Parenchymatous
(c)	Ground tissue in monocot stem	(iii)	Collenchymatous
(d)	Phloem parenchyma in monocot stem	(iv)	Sclerenchymatous

(1) (a) – (iv), (b) – (i), (c) – (iii), (d) – (ii)

(2) (a) – (i), (b) – (ii), (c) – (iv), (d) – (iii)

(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(4) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

4)

Consider the following statements (A–D) and select the option which includes all the **correct** ones :

(A) A leaf is said to be simple when its lamina is entire or when incised, the incision do not touch the mid rib or petiole. (B) A compound leaf has axillary bud in the axil of petiole. (C) Alternate phyllotaxy is found in *Calotropis* and Guava. (D) Reticulate venation is found in mango.

(1) A, B, D

(2) Only A

(3) A and B

(4) A and C

5) Study the sentences carefully and give the answer.
(A) When a flower can be divided into two equal radial halves through any plane, it is zygomorphic.
(B) Mustard, datura and chilli are examples of actinomorphic flowers.
(C) Pea, gulmohur, bean, *Cassia* are examples of zygomorphic flowers. (D) If a fruit is formed without fertilisation of the ovary, it is called geocarpic fruit.

(1) Both A and B are correct

(2) Both C and D are correct

(3) Both A and D are incorrect

(4) Both B and D are incorrect

6) (a) Leaves are arranged in acropetal or basipetal manner
(b) Leaves originate from shoot apical meristem
(c) Leaves develops at node and internodes.
Choose the correct statements :-

(1) a, b, c

(2) a, b

(3) b, c

(4) only b

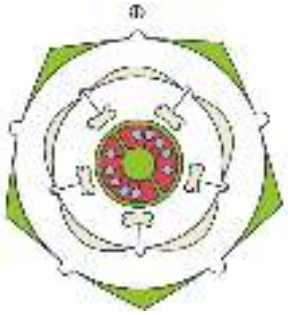
7) Correct floral formula of Makoi is :-

(1) ⊕ ♂♀ $K_{(5)}C_{(5)}A_5\underline{G}_{(2)}$

(2) ⊕ ♂♀ $K_{(5)}\overbrace{C_5A_5}G_{(2)}$

(3) ⊕ ♂♀ $K_{(5)}\overbrace{C_{(5)}A_5}\underline{G}_{(2)}$

(4) ⊕ ♂♀ $K_{(5)}\overbrace{C_{(5)}A_{(5)}}\underline{G}_{(2)}$



8) The following floral diagram belongs to which family ?

(1) Brassicaceae

(2) Fabaceae

(3) Liliaceae

(4) Solanaceae

9) **Assertion (A) :-** Both wind and water pollinated flowers are not very colourful and do not produce nectar.
Reason (R):- Wind and water pollinated flowers do not need to attract pollinating agents like insects.

(1) Both (A) & (R) are correct and (R) is correct explaination of (A)

(2) Both (A) & (R) are correct but (R) is not correct explaination of (A)

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

10) **Statement-I :** Pollination by water is quite rare in flowering plants & is limited to about 30 genera.
Statement-II : In most of water-pollinated species, pollen grains are protected from wetting by a mucilagenous covering.

(1) Only statement I is correct.

(2) Only statement II is correct

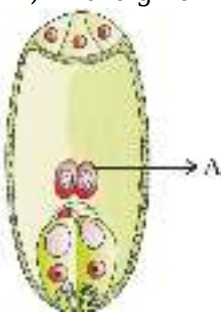
(3) Statement I & II both are correct.

(4) Statement I & II both are incorrect.

11) **Statement-1:** In all flowering plants, before fertilisation, ovary develops into fruit and ovules mature into seeds.
Statement-2: Events in sexual reproduction after the formation of zygote are called post fertilization events.

(1) Both statements are correct
(2) Statement-1 is correct and statement-2 is incorrect
(3) Statement-1 is incorrect and statement-2 is correct
(4) Both statements are incorrect

12) In the given figure, which of the following statements are **true** about the labelled region "A" :-



(a) Participate in double fertilization
(b) Participate in triple fusion
(c) Participate in embryo formation
(d) Participate in endosperm formation
Options :-

(1) a and c only
(2) b and d only
(3) a, b and d only
(4) a and b only

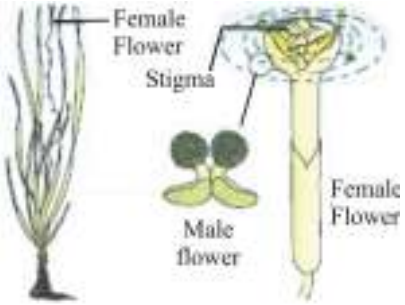
13) The pollen grain represents the :-

(1) Female gametophyte
(2) Male gamete
(3) Male gametophyte
(4) Female gamete

14) What will be the ploidy of the cells of the nucellus, megaspore mother cell, functional megaspore and female gametophyte respectively ?

(1) 2n, 2n, n, 2n
(2) 3n, 2n n, n
(3) 2n, n, 2n, n
(4) 2n, 2n, n, n

15) Given below is the diagram of pollination in plants. Which type of pollination is represented by this figure ?

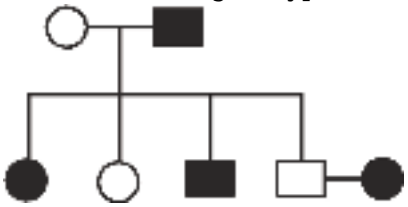


(1) Anemophily
(2) Entomophily
(3) Hydrophily
(4) Ornithophily

16) Select the odd one out with respect to chromosomal theory of inheritance.

(1) It was proposed by Sutton and Boveri
(2) Behaviour of chromosomes is parallel to behaviour of genes
(3) Chromosomes and genes occur in pairs in diploid and triploid cells, respectively
(4) The paired condition of both chromosomes as well as Mendelian factors are restored during fertilization

17) Find out the genotype of father and mother in the given pedigree chart :-



	Mother	Father
1	AA	Aa
2	Aa	Aa
3	AA	aa
4	aa	Aa

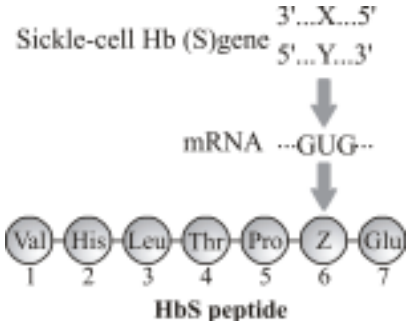
(1) 1
(2) 2
(3) 3
(4) 4

18) Which of the following statements is not true ?

(1) Two organism with the same genotype may have same phenotypes
(2) Two organisms with the same phenotype may have different genotypes
(3) A heterozygous organism always have the different phenotype as a homozygous organism

(4) A heterozygous organism has the same number of alleles for a given gene as a homozygous organism

19) Given below is the diagram related to sickle-cell anaemia, which is an autosomal recessive trait.



In this identify X, Y and Z :-

(1) X – CAC ; Y – GTG ; Z – Val

(2) X – CAC ; Y – GTG ; Z – Glu

(3) X – GTG ; Y – CAC ; Z – Val

(4) X – CAC ; Y – GTG ; Z – His

20) What will be the percentage of pure plants in F_2 generation of Mendel's monohybrid Cross-

(1) 12.5%

(2) 50%

(3) 6.25%

(4) 87.5%

21) Read the following statements regarding Mendelian inheritance and choose the correct option:
A. Mendel's experiments had small sample size which gave greater credibility to the data.
B. A true breeding line shows a stable trait inheritance and expression for several generations.
C. In a dissimilar pair of factors, one member of the pair dominates over the other.
D. A recessive parental trait is expressed only in its heterozygous condition.
E. Two alleles of a gene are located at same locus on homologous chromosomes.

(1) B alone is correct

(2) B, C and E are correct

(3) A and D are correct

(4) A, C and E are correct

22) The main enzyme of DNA replication is referred to as DNA dependent DNA polymerase because :-

(1) It uses a DNA template to catalyse the polymerisation of deoxynucleosides

(2) It uses RNA primer to catalyse the polymerisation of deoxynucleotides

(3) It uses a DNA template to catalyse the polymerisation of RNA primer

(4) It uses a DNA template to catalyse the polymerisation of deoxynucleotides

23) Which of the following statement is true about transcription ?

(1) Transcription is the process of copying genetic information from both the strand of a gene into RNA

(2) The principle of complementarity governs the process of transcription

(3) DNA of an organism get duplicated

(4) Total DNA of an organism is copied into RNA

24) Which of the following statement is incorrect ?

(1) The DNA in nucleoid is organised in large loops held by proteins

(2) Histone protein acquires charge depending upon the abundance of amino acids residues with charged side chains

(3) The packaging of chromatin at higher level requires additional set of proteins called as Non-histone chromosomal proteins

(4) Heterochromatin is transcriptionally active chromatin

25)

Which statement is incorrect for VNTR ?
(i) VNTR belongs to a class of satellite DNA referred to as microsatellite.
(ii) Copy number of VNTR varies from chromosome to chromosome in an individual.
(iii) The size of VNTR varies from 0.1 to 20 Kb.

(iv) The number of repeat of VNTR show very low degree of polymorphism.

(1) (i) and (ii)

(2) (i), (ii), (iii) and (iv)

(3) (ii) and (iii)

(4) (i) and (iv)

26) Meselson and Stahl experiment is continued to 2nd generation, then ratio of DNA band in centrifugation tube which has density
$N^{14}N^{15}$: $N^{14}N^{14}$: $N^{15}N^{15}$

(1) 1 : 1 : 0

(2) 1 : 1 : 1

(3) 0 : 1 : 1

(4) 1 : 2 : 1

27) Which of the following pair is correct about Human genome project ?

(1) EST Method → sequencing of only coding part

(2) Sequence Annotation Method → sequencing only of non-coding part

(3) Automated DNA sequencers → sequences only coding part

(4) Cloning Vectors → Ti and Ri plasmid

28) Read the following statements.
(A) In human, transcription and translation takes place in cytoplasm
(B) In humans, exons are interrupted by introns
(C) Intron as a segment of DNA coding for a polypeptide.

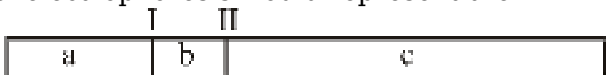
(D) The principle of complementarity governs the process of fat synthesis.
How many statements are incorrect ?

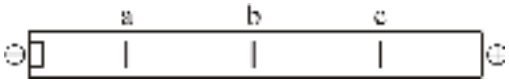
(1) one
(2) two
(3) three
(4) All

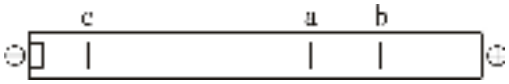
29) Severo ochoa enzyme is :-

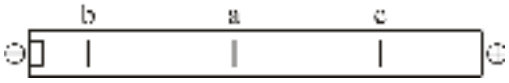
(1) Polyribonucleotide phosphorylase
(2) Polyribonucleotide dephosphorylase
(3) Polydeoxyribo nucleotide dephosphorylase
(4) Polydeoxyribo nucleotide phosphorylase

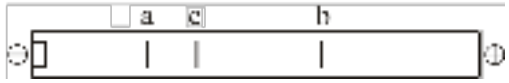
30) A segment of DNA has restriction sites I & II which create restriction fragments a, b and c. Which of the following arrangements produced by gel electrophoresis would represent the separation and correct position of these fragments ?



(1) 

(2) 

(3) 

(4) 

31) Which of the following statement is not correct about pyramid of biomass ?

(1) It may be upright or inverted.
(2) It is based on standing crop.
(3) A large standing crop is always required to support a small standing crop.
(4) In a forest ecosystem it is always upright.

32) Clear stratification is found in :–

(1) Tropical rain forest
(2) Deciduous forest
(3) Temperate evergreen forest
(4) Alpine forest

33) Productivity is :-

(1) Basal metabolic rate
(2) Rate of biomass production

(3) Daily metabolic rate

(4) Energy consumption rate during growth and reproduction

34)

In a simple food chain, the energy is lost in the form of heat. Which statement is correct :-
(a) Producers lose heat energy
(b) Herbivores lose heat energy

(c) Consumers lose heat energy

(1) All are wrong

(2) b & c is correct, a is wrong

(3) All are correct

(4) Only c is correct

35) Which of the following statement about GFC is incorrect ?

(1) Sun is the source of energy

(2) Begins with consumers

(3) Major conduit of energy flow in aquatic ecosystem

(4) Size of organism commonly increase at higher trophic level

36) Choose the incorrectly matched pair :-

(1) Proto-cooperation → Obligatory relationship

(2) Mutualism → Lichens

(3) Predation → Conduit of energy

(4) Amensalism → Negative interaction

37) What is/are **correct** about parasite ?
(a) Parasite can be host specific
(b) Biotic potential of parasite is high
(c) Parasite do not directly kill their hosts.
(d) Parasite and host tend to co-evolve

(1) a, b

(2) a, b, c

(3) a, b, c, d

(4) None

38) A bird introduced from another country becomes a serious pest due to :

(1) Better adaptation to new area.

(2) Increased sexual reproduction.

(3) Better nesting habitates.

(4) Absence of their natural predator

39) Identify the correct match from the column I, II and III.

	Column-I		**Column-II**		**Column-III**
1.	Mutualism	a.	(+, –)	i.	Visiting flamingoes and resident fishes in south american lakes
2.	Competition	b.	(+, 0)	ii.	Starfish pisaster in american pacific coast
3.	Predation	c.	(+, +)	iii.	Cattle egret birds and cattle
4.	Commensalism	d	(–, –)	iv.	Lichens

(1) 1 - d - iv, 2 - c - i, 3 - a - ii, 4 - b - iii

(2) 1 - c - iv, 2 - d - i, 3 - a - ii, 4 - b - iii

(3) 1 - c - iv, 2 - d - iii, 3 - a - ii, 4 - b - i

(4) 1 - d - iv, 2 - c - i, 3 - b - ii, 4 - a - iii

40) Which statement is **incorrect** regarding functions of predators ?

(1) Predators act as conduit for energy transfer across trophic levels.

(2) Predators over exploits its prey.

(3) Predators keep their prey populations under control.

(4) Predators maintain species diversity in a prey community.

41) **Assertion (A) :** Amazon forest is estimated to produce only 2% of the total oxygen in the earth's atmosphere.
Reason (R) : Amazon Rain forest is the very low biodiversity rich region.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) incorrect

42) **Assertion :** For many taxonomic groups, species inventories are more complete in temperate than in tropical countries.
Reason : Conventional taxonomic methods are not suitable for identifying species in tropics.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

43) Match the animal given in column-I with their place in column-II :-

Column-I		**Column-II**	
(i)	Dodo	(a)	Africa
(ii)	Quagga	(b)	Russia

(iii)	Thylacine	(c)	Mauritius
(iv)	Steller's sea cow	(d)	Australia

(1) i-a, ii-c, iii-b, iv-d

(2) i-d, ii-c, iii-a, iv-b

(3) i-c, ii-a, iii-b, iv-d

(4) i-c, ii-a, iii-d, iv-b

44) **Statement-I :** Alien species invasion is the most important cause driving animals and plant to extinction.
Statement-II : Species extinctions in the last 500 years (Steller's Sea cow, passenger pigeon) was due to Habitat loss and fragmentation.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

45) **Statement-I :** National park, wildlife sanctuaries and biosphere reserves are the examples of Ex-situ conservation.
Statement-II : Exploring molecular, genetic and species level diversity for the products of economic importance known as bioprospecting.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

46) Fill in the blanks.
" Enzyme isolated from ____A____ *Thermus aquatics* which remain active during ____B____ in PCR.

	(A)	**(B)**
(1)	Virus	High temperature
(2)	Bacteria	Low temperature
(3)	*Mycoplasma*	High temperature
(4)	Bacteria	High temperature

(1) 1

(2) 2

(3) 3

(4) 4

47) What is the meaning of elution in recombination DNA technology.

(1) Insertion of desired DNA into the host cell

(2) The separated band of DNA are cut out from the gel and extracted from the gel piece

(3) Production of injectable hepatitis-B vaccine
(4) Amplification of DNA fragment by PCR

48) Which of the following is not related with downstream processing ?

(1) Expression
(2) Separation
(3) Purification
(4) Preservation

49) Which of the following is incorrectly matched ?

(1) *Agrobacterium tumefaciens* → Ti plasmid
(2) Retrovirus → Suitable for animal cell
(3) Biolistic method → Suitable for animal cell
(4) Divalent cation → Suitable for bacterial cell

50) Select the incorrect statement :-

(1) PCR is in-vivo reaction
(2) In PCR amplification is achieved by the use of a thermostable DNA polymerase
(3) Thermostable enzyme is obtained from bacteria *Thermus aquaticus*
(4) Taq DNA polymerase enzyme extends the primers using nucleotides provided in the reaction

51) How many statements are correct in the following-
(i) Insulin is a global protein used for treating diabetes.
(ii) Insulin is the first hormone prepared by genetic engineering.
(iii) T-DNA is present in the plasmid of *Agrobacterium tumefaciens.*
(iv) Taq polymerase is a thermosensitive enzyme.

(1) One
(2) Three
(3) Only two
(4) All

52) Read following statements carefully :
(A) Main challenges for production of insulin using rDNA techniques was to assemble into mature form.
(B) Cry I Ac and Cry I Ab control the cotton bollworms.
(C) Bt toxins are insect group specific.
(D) C-Peptide is absent in mature insulin.
How many above are correct ?

(1) Two
(2) Three
(3) One
(4) Four

53) **Statement-I :-** Totipotency is the capacity to generate a whole plant from any living cell of the plant.
Statement-II :- Producing thousand of plants through tissue culture is called micro-propagation.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

54) Match List-I with List-II.

	List-I		**List-II**
(A)	ADA	(I)	*Bacillus thuringiensis*
(B)	PCR	(II)	Gene therapy
(C)	Bt-cotton	(III)	Pomato
(D)	Somatic hybridisation	(IV)	Detect HIV infection

(1) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(2) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(4) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

55) **Statement-I :-** *Flavr Savr* is a genetically modified tomato, which remains fresh and retains its flavour much longer than the normal tomato due to blocking of synthesis of fruit softening enzyme polygalacturonase.
Statement-II :- Useful biological products can be produced by introducing the portion of DNA which codes for particular product into transgenic animals.

(1) **Statement I** & **Statement II** both are correct.

(2) **Statement I** is correct & **Statement II** is incorrect.

(3) **Statement I** is incorrect & **Statement II** is correct.

(4) Both **Statement I** & **Statement II** both are incorrect.

56) Which of the following statements are correct regarding the process of RNA interference-
(i) The method involves silencing of a specific mRNA due to a complementary dsRNA molecules.
(ii) This is used to prevent the infection of protozoans
(iii) It takes place in some eukaryotic and all prokarytoic organisms as a method of cellular defense.
(iv) It is novel strategy to produce pest resistant plants.

(1) (i), (ii) and (iv)

(2) (ii) and (iii)

(3) (i) and (iv)

(4) (i), (ii) and (iii)

57) Given below are some steps of formation of insulin by genetic engineering :-
A. Chain A and B extracted separately.

B. Introduced gene of chain A and B transfer in plasmids of *E. coli* to produce insulin chains
C. Chain A and B combined by creating disulfide bonds
D. Chain A and B were produced separately
E. Prepared two DNA sequence corresponding to A and B chains
Choose the correct sequence of steps from the options given below :-

(1) C-E-D-A-B
(2) E-B-D-A-C
(3) B-D-E-A-C
(4) E-C-A-D-B

58) Given below are two statements one is labelled as assertion and other is labelled as reason.
Assertion : Baculoviruses are pathogen that attack insects and other arthropods.
Reason : The majority of Baculoviruses used as biocontrol agents are in the genus Ribonucleopolyhedroviruses.

(1) Both assertion and reason are true and reason is not correct explanation of assertion.
(2) Assertion is true and reason is false
(3) Assertion is false and reason is true.
(4) Both assertion and reason are true and reason is correct explanation of assertion.

59) **Assertion (A) :** *Lactobacillus* and lactic acid bacteria (LAB) grow in milk and convert it to curd.
Reason (R) : During growth LAB produce acids that coagulate and partially digest the milk protein.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

60) Match the column-I with column-II and select correct options :-

	Column-I		**Column-II**
(A)	*Aspergillus niger*	(i)	Dough
(B)	*Trichoderma polysporum*	(ii)	Organic acid
(C)	*Saccharomyces cerevisiae*	(iii)	Immunosupressive
(D)	*Propionibacterium sharmanii*	(iv)	Cheese

(1) A-ii, B-iii, C-i, D-iv
(2) A-ii, B-iii, C-iv, D-i
(3) A-iii, B-ii, C-i, D-iv
(4) A-i, B-ii, C-iii, D-iv

61) Match the column and choose the correct option ?

I		II		III	
A.	*Homo habilis*	i	1400 cc	a.	First man to use fire
B.	*Homo erectus*	ii	650-800 cc	b.	Cave painting
C.	Neanderthal man	iii.	1650 cc	c.	First tool maker
D.	Cromagnon man	iv	900 cc	d.	Buried their dead

(1) A-iv-a, B-ii-c, C-i-b, D-iii-b

(2) A-ii-c, B-i-a, C-iii-b, D-iv-b

(3) A-ii-c, B-iv-a, C-i-d, D-iii-b

(4) A-iv-b, B-ii-c, C-iii-b, D-iii-b

62) **Assertion :-** Divergent evolution is based on adaptive radiation.
Reason :- Wings of butterfly & birds show divergent evolution.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

63)

Match **List-I** with **List-II**.

	List-I		List-II
A.	Gene pool	I.	Stable within a generation
B.	Genetic drift	II.	Change in gene frequency by chance
C.	Gene flow	III.	Transfer of genes into or out of population.
D.	Gene frequency	IV.	Total number of genes and their alleles

Choose the **correct** answer from the options given below :

(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-II, C-III, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

64) Given below are two statement.
Statement-I :- Fossils are remain of hard parts of life forms found in rocks.
Statement-II :- The geological history of earth closely correlates with the biological history of earth.

(1) Statement-I is true but statement-II is false.
(2) Statement-I is false but statement-II is true.
(3) Both statement-I and statement-II are true.
(4) Both statement-I and statement-II are false.

65) Which of the following statement is correct ?

(1) The rate of disappearance of new forms is linked to the life cycle.
(2) Fitness is the end result of the ability to adapt and get selected by nature.
(3) Branching descent is the key concept of lamarck.
(4) According to Darwin, evolution is driven by use and disuse by organs.

66) Presence of gills in the tadpole of frog indicated that

(1) Fishes were amphibious in the past
(2) Fishes evolved from frog-like ancestors
(3) Frogs will have gills in future
(4) Frogs evolved from gilled ancestors

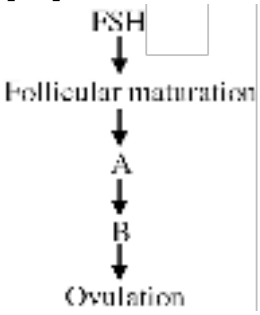
67) Darwin's finches are example of :-

(1) Biogeographical evidence
(2) Paleontological evidences
(3) Embryological evidences
(4) Analogous organs

68) Hugo de varies worked on-

(1) *Pisum sativum*
(2) *Evening primrose*
(3) *Lathyrus odoratus*
(4) *Arachis hypogea*

69) The menstrual cycle is regulated by interaction of hypothalamus, pituitary and ovarian hormones. Follicular growth and estrogen secretion prepare endometrium, while ovulation marks



the transition from proliferative to secretory phase. What are A and B respectively ?

(1) Progesterone rise, Estrogen fall
(2) Estrogen rise, progesterone rise

(3) Progesterone rise, LH rise

(4) Estrogen rise, LH rise

70) **Assertion (A):-** The second meiotic division of the secondary oocyte completes during fertilisation.
Reason (R) :- The second meiotic division is unequal and results in the formation of a second polar body and a diploid ovum.

(1) A is false but R is true

(2) A is true but R is false

(3) Both A and R are true and R is not the correct explanation of A

(4) Both A and R are true and R is the correct explanation of A

71) Match the following :

	Column 1		**Column 2**
(A)	Testis	(i)	Implantation
(B)	Uterus	(ii)	Accessory gland
(C)	Prostate	(iii)	External genitalia
(D)	Vulva	(iv)	Gametogenesis

(1) A-iv, B-i ,C-ii, D-iii

(2) A-iii, B-i ,C-ii, D-iv

(3) A-iv, B-iii ,C-iv, D-i

(4) A-ii, B-iv ,C-i, D-iii

72) Match column-I with column-II.

	Column-I		**Column-II**
A.	LH	I.	Maintenance of endometrium.
B.	FSH	II.	Ovulation
C.	Progesterone	III.	Parturition
D.	Oxytocin	IV.	Follicle development

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-II, C-III, D-IV

73) Read the following statements and find out the incorrect one.

(1) Only one sperm fertilizes the human egg.

(2) The secretion of acrosome help the sperm enter into cytoplasm of ovum through zona pellucida.

(3) All copulations lead to fertilization

(4) There is no pregnancy without implantation.

74) The layer which is attached with the endometrium of uterus during process of implantation is :-

(1) Corona radiata
(2) Trophoblast
(3) Zona pellucida
(4) Inner cell mass

75) Acrosome of sperm is derived from which cell organelles ?

(1) Mitochondria
(2) Nucleus
(3) Golgi apparatus
(4) Ribosome

76) Match the column:

	Column-I		**Column-II**
(a)	Chorionic villi	(i)	Connects embryo to placenta
(b)	Placenta	(ii)	Finger like projection on trophoblast
(c)	Umbilical cord	(iii)	Functional and structural unit between embryo and mother

(1) a-(ii), b-(iii), c-(i)
(2) a-(i), b-(iii), c-(ii)
(3) a-(ii), b-(i), c-(iii)
(4) a-(iii), b-(i), c-(ii)

77) Uterine contractions called foetal ejection reflex for parturition is induced by :

(1) Oxytocin
(2) Fully developed foetus
(3) Placenta
(4) Both 2 and 3

78) If a female undergoes tubectomy then which of the following event will not takes place?
(A) Formation of zygote
(B) Secretion of estrogen from ovaries
(C) Cyclic changes in the uterine endometrium
(D) Fertilisation of secondary oocyte with sperm
(E) Release of FSH and LH from pituitary
Choose the correct answer from the options given below:

(1) A and B only

(2) B and C only

(3) D and E only

(4) A and D only

79) **Statement-I :-** If male is unable to inseminate then IUI is performed. **Statement-II :-** IUI can also be done if sperm count is less than normal.

(1) Statement-I & II both are correct

(2) Statement-I & II both are incorrect

(3) Statement-I is correct but II is incorrect

(4) Statement-I is incorrect but II is correct

80) MTP (Medical Termination of Pregnancy) with oral pills is considered relatively safe upto-

(1) 15-18 weeks of pregnancy

(2) 20-24 weeks of pregnancy

(3) 7-9 weeks of pregnancy

(4) 28-30 weeks of pregnancy

81) Late detection and improper treatment of STIs leads to:

(1) Pelvic inflammatory diseases

(2) Abortions, still birth, ectopic pregnancies

(3) Infertility or cancer of reproductive tract

(4) All of these

82) Salmonella typhi generally enter the small intestine through contaminated food and water and migrate to other organs through blood.
Given below are two statement based on the above paragraph.
Statement-I : Typhoid is a bacterial disease characterised by sustained high grade fever, weakness, stomach pain headache and loss of appetite etc.
Statement-II : Typhoid fever could be confirmed by widal test. Intestinal perforation and death may occur in severe cases.
In the light of the above statement choose the most appropriate answer from the options given below.

(1) Both statement-I and II are correct.

(2) Both statement-I and II are incorrect.

(3) Statement-I is correct but-II is incorrect.

(4) Statement-I is incorrect but-II is correct.

83) Given below are two statements:
Assertion A : *Plasmodium falciparum* causes malignant malaria.
Reason R : *Plasmodium* is transmitted to humans through the bite of infected female *Anopheles* mosquitoes.
In the light of the above statements, choose the most appropriate answer from the options given

below.

(1) **A** is correct but **R** is not correct.

(2) **A** is not correct but **R** is correct.

(3) Both **A** and R are correct and **R** is the correct explanation of **A**.

(4) Both **A** and R are correct but **R** is NOT the correct explanation of **A**.

84) Read the following statements : -
(A) Skin on our body is the main barrier which allow entry of the micro-organisms.
(B) Acid in stomach and saliva in mouth prevent microbial growth
(C) Innate immunity is non-specific type of defence.
(D) Innate immunity is characterised by memory.
How many of the above statements are incorrect?

(1) Two

(2) Three

(3) One

(4) Four

85) **Statement-I :** Normal cells show a property called contact inhibition by virtue of which contact with other cells and inhibit their uncontrolled growth.
Statement-II : Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents.
In the light of the above statements choose the correct answer from the options given below.

(1) Statement-I and Statement-II both are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Statement-I and Statement-II both are incorrect.

86) The __A_____ themselves do not secrete antibodies but help the B-cells to produce them, and each antibody molecule has ___B__ polypetide chains. Find out correct answer about A and B from given below options.

(1) A- Macrophage, B-6

(2) A- Neutrophils, B-8

(3) A - T-cells, B-4

(4) A - Natural killer cell, B-4

87) Match the list-I with list-II and choose correct answer about them:

	List-I		**List-II**
(A)	Cytokine barriers	(i)	Uses strong magnetic field and non-ionising radiations.
(B)	Graft rejection	(ii)	Thymus and bone marrow

(C)	Primary lymphoid organs	(iii)	Cell mediated immunity
(D)	MRI	(iv)	Interferons

(1) A – i, B – ii, C – iii, D – iv

(2) A – iv, B – iii, C – ii, D – i

(3) A – iii, B – iv, C – i, D – ii

(4) A – ii, B – i, C – iv, D – iii

88) Which of the following is not an autoimmune disease?

(1) Allergy

(2) Rheumatoid Arthritis

(3) Hashimoto's disease

(4) Myasthenia gravis

89)

Erythroxylum coca is a native plant of -

(1) South Africa

(2) South America

(3) Australia

(4) Asia

90) Active immunity is :

(1) Fast and short-lasting

(2) Slow and long-lasting

(3) Due to transfer of antibodies

(4) Without memory formation

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	1	2	3	4	2	2	2	2	3	1	3	2	2	1	1	2	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	4	3	1	4	2	1	2	4	1	1	3	1	4	1	3	1	2	2
Q.	41	42	43	44	45															
A.	2	1	3	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	1	2	2	4	2	3	1	3	2	2	2	3	2	3	3	2	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	2	4	4	2	4	1	2	3	3	3	2	1	1	1	4	2	3
Q.	86	87	88	89	90															
A.	1	2	4	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	3	1	3	4	3	4	1	3	3	3	3	4	3	3	4	3	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	2	4	4	1	1	3	1	2	3	1	2	3	2	1	3	4	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	2	4	4	2	1	3	1	2	2	1	1	1	3	2	2	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	3	2	4	1	2	4	2	1	3	3	2	3	1	4	4	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	1	4	1	1	3	2	1	2	2										

SOLUTIONS

PHYSICS

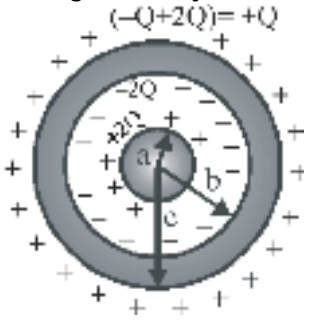
1)

After redistribution of charge by induction,
Net charge on inner surface of shell = –2Q

Charge density = $\frac{-2Q}{4\pi b^2}$

Net charge on outer surface of shell = –Q + 2Q = +Q

Charge density = $\frac{+Q}{4\pi c^2}$



Final Answer: Option (1).

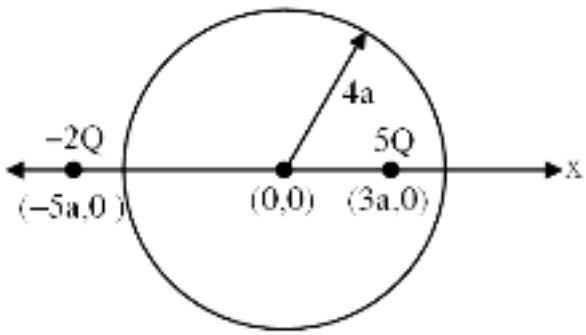
2) $\frac{F_1}{F_2} = \frac{|-10 \times 20|}{5 \times 5}$

$\frac{F_1}{F_2} = \frac{|-8|}{1} = \frac{8}{1}$

3)

Field lines of q_1 passes through surface of hemisphere one time.
Field lines of q_2 passes through surface of hemisphere two time so net flux due to q_2 is zero.
Net flux due to q_1 is non zero.

4)



5Q charge is inside the spherical region

flux through sphere = $\frac{5Q}{\varepsilon_0}$

5) Electric field inside conductor is always zero in electrostatic condition & if electric field is zero then all point will have same potential

as $\Delta V = -\int \vec{E} \cdot d\vec{r}$

So if $\vec{E} = 0$ then $\int \vec{E} \cdot d\vec{r} = 0$ then $\Delta V = 0$

So there will be no potential difference, all point will have same potential.

6) Potential difference between A and B

$V_A - V_B = 1 \times 1.5$

$\Rightarrow V_A - 0 = 1.5V \Rightarrow V_A = 1.5V$

Potential difference between B and C

$V_B - V_C = 1 \times 2.5 = 2.5V$

$\Rightarrow 0 - V_C = 2.5\ V \Rightarrow V_C = -2.5\ V$

Potential difference between C and D

$V_C - V_D = -2V \Rightarrow -2.5 - V_D = -2$

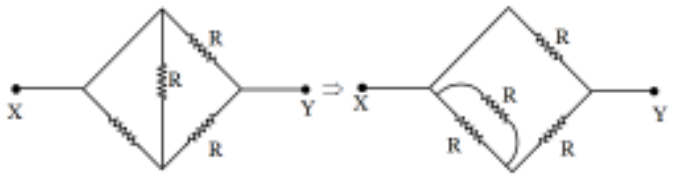
$\Rightarrow V_D = -0.5V.$

7)

for balancing $\frac{P}{Q} = \frac{R}{\frac{S(6)}{S+6}} \Rightarrow \frac{2}{2} = \frac{2}{\frac{S(6)}{S+6}}$

$\frac{6S}{6+S} = 2 \Rightarrow S = 3\Omega$

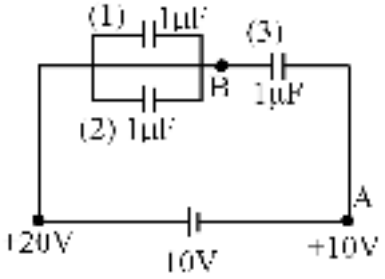
8) Equivalent ckt.



$R_{equation} = \frac{\frac{3R}{2} \times R}{\left(\frac{3R}{2} + R\right)} = \left(\frac{3R}{5}\right)$

9) $R_s = \frac{I_g R_g}{I - I_g} = \frac{10^{-2}(50)}{(1 - 0.01)} = \frac{50}{99}\Omega$

10)



Voltage across capacitor (3)

$V_3 = \frac{2}{3} \times 10 = \frac{20}{3}$

Applying KVL from A

$10\ V + \frac{20}{3} = V_B$

$V_B = \frac{50}{3} V$

11) Net EMF of 9v & 18v cell

$E_{Net} = \frac{18 \times 2 + 9 \times 4}{2 + 4} = \frac{72}{6} = 12v$

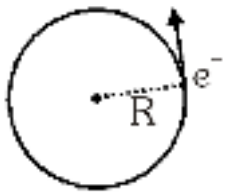
$\Rightarrow v_c = 12 - 9 = 3v$

12) $\vec{B} = B \sin\theta \hat{i} + B \cos\theta (-\hat{j})$

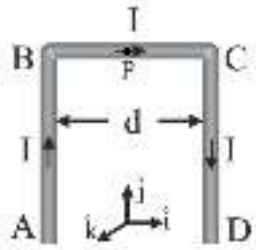
$= B\left(\frac{y}{r}\hat{i} - \frac{x}{r}\hat{j}\right) = \frac{\mu_0 I}{2\pi r}\left(\frac{y}{r}\hat{i} - \frac{x}{r}\hat{j}\right)$

$= \frac{\mu_0 I(y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}, r = \sqrt{x^2 + y^2}$

13) $M = iA = \frac{e\pi R^2}{T}$



14)



$B_{AB} = \frac{\mu_0 I}{4\pi\left(\frac{d}{2}\right)}(-\hat{k}) = \frac{\mu_0 I}{2\pi d}(-\hat{k})$

Using right Hand Thumb Rule, the magnetic field due to CD at P is directed into the page $(-\hat{k})$

$B_{CD} = \frac{\mu_0 I}{4\pi\left(\frac{d}{2}\right)}(-\hat{k}) = \frac{\mu_0 I}{2\pi d}(-\hat{k})$

$B_{net} = B_{AB} + B_{BC} + B_{CD}$

$$B_{net} = \frac{\mu_0 I}{2\pi d}\hat{k} + 0 + \frac{-\mu_0 I}{2\pi d}\hat{k}$$

$$B_{net} = \frac{-\mu_0 I}{\pi d}\hat{k}$$

Final Answer: Option (2)

15) $B = \frac{\mu_0 i}{4r} = 10^{-7} \times \frac{\pi \times 10}{20 \times 10^{-2}} = B = 5\pi\mu T$

16)

A magnetic field exerts a force perpendicular to the velocity of a charged particle, altering its direction (momentum) but not its speed (kinetic energy).

17)

$$B_{centre} = \frac{\mu_0(I)}{2r} = \frac{\mu_0\left(\frac{q}{t}\right)}{2r} = \frac{\mu_0\left(\frac{Ne}{t}\right)}{2r} = \frac{\mu_0\left(\frac{1\times e}{T}\right)}{2r} = \frac{\mu_0\left(\frac{e}{T}\right)}{2r} = \frac{\mu_0(\nu e)}{2r} = \frac{\mu_0 \nu e}{2r}$$

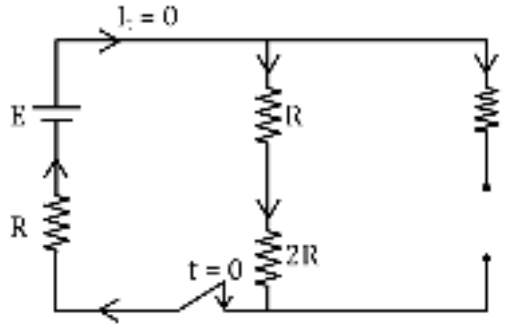
18) $e = M\frac{di}{dt} = 0.005 \times \frac{d}{dt}(I^0 \sin\omega t)$

$= 0.005 \times I_0\omega\cos\omega t$

$\therefore e_{max} = 0.005 \times 10 \times 100\pi = 5\pi$

[$\because \cos\omega t = 1$]

19) **Solution:** As we know, from EMI, just after closing the switch i.e. at t =0, the inductor "L" behaves like an open circuit.



therefore effective circuit will be like,

$$I_t = \frac{E}{4R}$$

Final Answer: $\frac{E}{4R}$

20) $e = \frac{d\phi}{dt} = \frac{d}{dt}(\pi r^2 B) = 2\pi r B\frac{dr}{dt}$

$\Rightarrow \quad r = \dfrac{e}{\left(2\pi B\frac{dr}{dt}\right)}$

$\Rightarrow \quad r = \dfrac{10^{-6}}{2\pi \times 10^{-3} \times 10^{-2}} = \dfrac{5}{\pi}cm$

21) p.f. = $\cos\phi = \dfrac{1}{\sqrt{2}} = \dfrac{R}{Z}$

$\Rightarrow Z = \sqrt{2}R \Rightarrow Z = 10\sqrt{2}\Omega$

Here $Z = \sqrt{R^2 + (X_C - X_L)^2}$

where X_L = = ωL = (100)(0.1) = 10 Ω

$200 = 100 + (X_C - 10)^2$

$\Rightarrow X_C = 20\Omega$

$\Rightarrow \dfrac{1}{100C} = 20$

$\Rightarrow$ C = 500 µF

22)

2πf = 75π

2f = 75

current becomes zero two times in one cycle.

23)

Given that,

$E = 3.1\ NC^{-1} \times \cos[1.8\ rad\ m^{-1})y + (5.4 \times 10^8\ rad\ s^{-1})]t\hat{i}$ .....(i)

Comparing (i) with the equation $E = E_0 \cos(ky + \omega t)$

We get, $k = 1.8\ rad\ m^{-1}$,

$E_0 = 3.1\ NC^{-1}$, $c = 3 \times 10^8\ ms^{-1}$, $\omega = 5.4 \times 10^8\ rad\ s^{-1}$

Now, $\lambda = \dfrac{2\pi}{k} = \dfrac{2 \times 22}{1.8 \times 7} = 3.5$ m

24) (A)→(S); (B)→(Q); (C)→(P); (D)→(R)

Gauss's law in electrostatics $\phi_E = \oint_s \vec{E}.\vec{ds} = \dfrac{q}{\varepsilon_0}$

where, ϕ_E is electric flux and $\vec{ds}$ is a differential area vector on the closed surface.

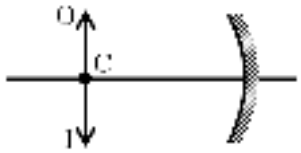
Gauss's law for magnetism $\oint_s \vec{B}.\vec{ds} = 0$ i.e., net magnetic field of a closed surface is always zero.

Faraday's law of electromagnetic induction $\oint_s \vec{E}.\vec{d\ell} = -\dfrac{d\phi_B}{dt}$

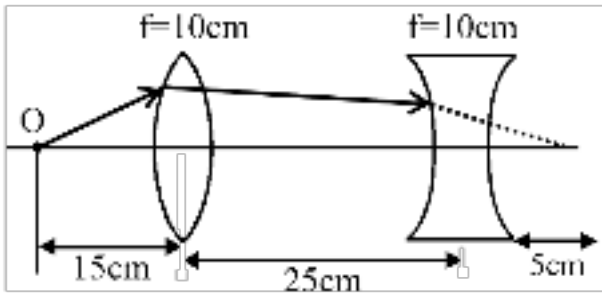
where $\dfrac{d\phi_B}{dt}$ is rate of change of magnetic flux through the circuit.

25)

δ = i + e – A i – e = 20°
40 = i + e – 60 i = 60°
i + e = 100 e = 40°

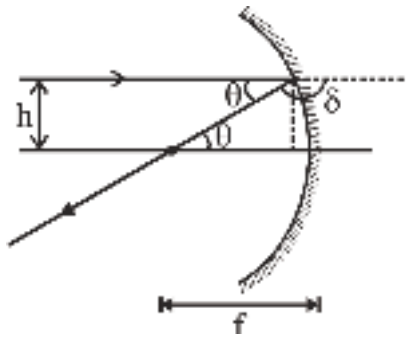
26) 

Distance between O & I = 0°

27) 

u = –15cm
f = 10 cm
v = 30 cm

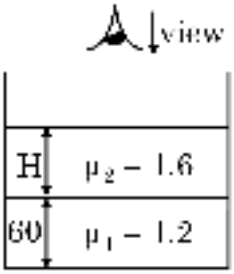
u = 5 cm
f = –10 cm
$v = \frac{uf}{u+f} = \frac{5 \times -10}{5-10}$
v = 10 cm

28) 

θ ~ $\tan\theta = \frac{h}{f}$

 $\delta = 180° - \frac{h}{f}$

29)



y = apparent depth of bottom

$$\frac{y}{1} = \frac{H}{1.6} + \frac{60}{1.2}$$

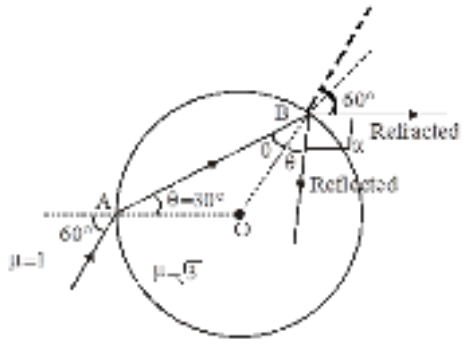
Shift = 40

H + 60 – y = 40

H + 60 – $\frac{H}{1.6} - \frac{60}{1.2}$ = 40

$\frac{6}{16}H = 30$ ⇒ H = 80 cm **OR** Shift = $d_{Acc} - d_{App}$

40 = (60 + H) – $\left[\frac{60}{1.2} + \frac{H}{1.6}\right]$



30)

By Snell's law at A :

$1 \times \sin 60° = \sqrt{3} \times \sin\theta \Rightarrow \theta = 30°$

So at B → θ + 60° + α = 180

30° + 60° + α = 180° ⇒ α = 90

31) $\beta = \frac{\lambda D}{d}$, which is independent of m.

32) $\Delta y = 3\frac{\lambda D}{d} - \frac{\lambda D}{2d}$

$\Delta y = \left(3 - \frac{1}{2}\right)\beta$

$= \frac{5}{2}\beta$

$= \frac{2.5 \times 480 \times 10^{-9} \times 2}{3 \times 10^{-3}}$

= 0.8 mm

33) Intensity, $I = A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi$

If $\Delta\phi = \frac{\pi}{3}$ then

$I_1 = A_1^2 + A_2^2 + 2A_1A_2\left(\frac{1}{2}\right) = A_1^2 + A_2^2 + A_1A_2$

If $\Delta\phi = \frac{2\pi}{3}$, then

$I_2 = A_1^2 + A_2^2 + 2A_1A_2\left(-\frac{1}{2}\right) = A_1^2 + A_2^2 - A_1A_2$

$\Rightarrow A_1^2 + A_2^2 = \frac{I_1 + I_2}{2}$ & $2A_1A_2 = I_1 - I_2$

Therefore

$I_{max} = (A_1 + A_2)^2 = A_1^2 + A_2^2 + 2A_1A_2$

$= \left(\frac{I_1 + I_2}{2}\right) + I_1 - I_2 = \frac{3I_1 - I_2}{2}$

34) Angular width $\propto \frac{\lambda}{d}$

$\frac{\theta_0}{0.3\theta_0} = \frac{\frac{6000\text{Å}}{d}}{\frac{\lambda}{d}}$

$\Rightarrow$

$\Rightarrow \lambda = 1800$ Å

35) Required reaction is $Cu^{64} \rightarrow Cu^{63} + n^1$

Q value = (63 × 8.75 – 64 × 8.73)

= 551.25 – 558.72 = –7.47 MeV

36) $\frac{1}{\lambda} = R\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$

For lyman series

$\frac{1}{\lambda_{min}} = R$ ...(1)

For Balmer series

$\frac{1}{\lambda'_{min}} = \frac{R}{4}$ ....(2)

$\lambda'_{min} = 4\lambda = 3648$ Å

37) E = nhv $\Rightarrow \nu \propto \frac{1}{n} \Rightarrow \frac{n_1}{n_2} = \frac{\nu_2}{\nu_1}$

38)

(A) True : Existence of threshold frequency show photons carry quantized energy.

(B) True : Kinetic energy depends only on frequency, not intensity, proving particle nature.

(C) True : Instantaneous emission above threshold frequency supports photon interaction.

(D) Not directly related to quantum nature of light, but rather to quantization of charge (already known).
Thus, the correct statements supporting quantum nature of light are A, B and C.

39) $\lambda \propto \frac{1}{\sqrt{V}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}} \Rightarrow \frac{1\text{Å}}{\lambda_2} = \sqrt{\frac{600}{150}} = 2$

$\lambda_2 = \frac{1}{2}$Å

$\lambda_2 = 0.5$Å

40)

In F.B. barrier height decreases & In R.B. barrier height increases.

41)

From the given waveform following truth table can be formed

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

This belongs to NOR gate

42) **Statement I :** A photodiode is a p – n junction diode designed to operate in reverse bias. In this configuration, a reverse voltage is applied across the junction, which increases the width of the depletion region. **Statement II :** When light falls on the diode, electron-hole pairs are generated with in or near the depletion region. These minority charge carriers are swept across the junction by the electric field, creating a measurable current. As minority current is small so fractional change in current is easily measurable. The change in this reverse current is proportional to the intensity of incident light, making the device useful for light detection. Both statements are correct

43) $R = \frac{V}{I} = \tan 53^\circ = \frac{4}{3}\Omega$; radius $r = 10^{-3}$ m

$\frac{4}{3} = \frac{\rho \ell}{A} \Rightarrow \rho = \frac{4}{3}\frac{A}{\ell} = \frac{4}{3}\frac{\left(3.14 \times 10^{-6}\right)}{(0.5)}$

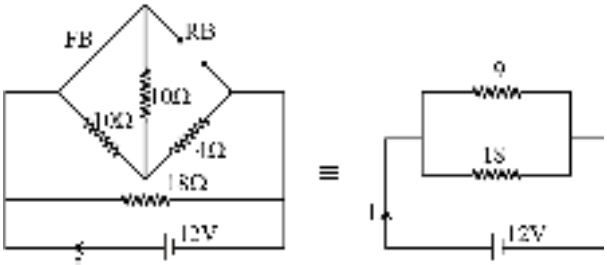
$\rho \approx 8 \times 10^{-6}$ Ω-m

44) The question asks you to match terms related to the hall-deflection method for determining the resistance and figure of merit of a galvanometer.

You need to pair each term from List-I with its correct description or context from List-ll
• Current Sensitivity. Thus deflection produced per unit current in the galvanometer.
• Galvanometer Figure of Merit. The current required to produce unit deflection in the galvanometer.
• Half-Deflection Method. A technique to determine resistance and figure of merit-by-halving the deflection using a shunt.
• Series and Parallel Connections High resistance is used in series, shunt in parallel to the galvanometer.
The correct matching is : a (Figure of merit K) - S(current per unit deflection).
B(Current sensitivity SI) - T(deflection per unit current).
C(Deflection is made half by using resistances) - R(in shunt/parallel).
D(High resistance box) - P(in series)

45)

Forward biased → zero resistance (short circuit)
Reverse biased → infinite resistance (open circuit)
equivalent resistance calculated below



$$R = \frac{10\Omega \times 10\Omega}{10\Omega + 10\Omega} = 5\Omega$$

5Ω and 4Ω are in series, therefore 5Ω + 4Ω = 9Ω
Now, 9Ω and 18Ω are in parallel, therefore

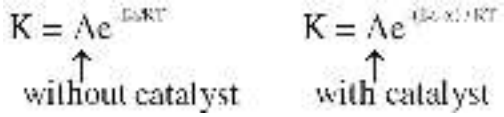
$$R_{eq} = \frac{9\Omega \times 18\Omega}{9\Omega + 18\Omega} = 6\Omega, \; R_{eq} = \frac{V}{I}$$

$$i = \frac{12}{6} = 2A$$

CHEMISTRY

46)

A → B
r = K[A]
(a) False – In $K = Ae^{-E_a/RT}$, Pre-exponential factor 'A' does not depend on catalyst
(b)True – As on increasing temperature in absence of catalyst rate will be more as Ea will be reduced in case of catalyst.



(c) False – Catalysed reaction will have reduced energy of activated complex.
(d) False - It may be or may not be elementary but the order is 1

47)

Reaction $A + B \rightarrow C$

Mechanism (I) $A + B \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} I$

(II) $I \xrightarrow{K_3} C$

From step (I) : $\frac{K_1}{K_2} = \frac{[I]}{[A].[B]}$

$\Rightarrow [I] = \frac{K_1}{K_2}[A].[B]$ ..... (1)

From step (II) : Rate = $K_3[I]$ ..... (2)

From eq. (1) & (2)

Rate = $\left(K_3.\frac{K_1}{K_2}\right)$ [A].[B]

Overall rate constant = K = $K_3.\frac{K_1}{K_2}$

$K = 0.01\ sec^{-1} \times \frac{2 \times 10^3}{10^3\ sec^{-1}}$ lit mol^{-1} sec^{-1}

$\boxed{K = 0.02 \frac{litre}{mol} sec^{-1}}$

48) $\frac{(ROD)CaCO_3}{Mw} = \frac{(ROF)CO_2}{Mw}$

$\frac{240}{100} = \frac{(ROF)CO_2}{Mw}$

$(ROF)CO_2 = \frac{240}{100} \times 44 = 105.6$ g L^{-1} s^{-1}

49)

	A(g)	→	2B (g)	+	C(g)
t = 0	100		0		0
t=t	100–x		2x		x

100 + 2x = 200

2x = 100

x = 50

$t_{1/2}$ = 100 min.

50) $E_{cell} = 0.0591\ \ pH_1 - pH_2 1$

For a concentration cell compound of two hydrogen electrodes, the maximum emf is obtained when the difference in pH between the two solutions is maximized. One electrode is standard ($[H^+]$ = 1M, pH = 0). To maximize E_{Cell}, the other electrode must have the highest possible pH (lowest acidity). Comparing 0.1M solutions : HCl and H_2SO_4 are strong acid (pH = 1 and < 1) H_3PO_4 ($K_a = 10^{-3}$) is a moderately weak acid, and CH_3COOH ($K_a = 2 \times 10^{-5}$) is the weakest acid among the choices. Since CH_3COOH has the smallest K_a, it yields the lowest $[H^+]$ and thus the highest pH, leading to the largest pH difference and maximum emf.

51)

Higher standard E° → stronger oxidizing agent.
An oxidant A (reduced form has E°A) will oxidize B (whose reduction couple has E°B) if E°(A) > E°(B) so that E°cell = E°(A) − E°(B) is positive.
For Ce^{4+} vs PbO_2/Pb^{2+}: E°cell = 1.61 − 1.455 = +0.155 V → spontaneous, so Ce^{4+} oxidises Pb^{2+} to PbO_2.
For MnO_4^- vs PbO_2/Pb^{2+}: E°cell = 1.51 − 1.455 = +0.055 V → spontaneous, so MnO_4^- oxidises Pb^{2+} to PbO_2.
For H_2O_2 vs Mn^{2+}/MnO_4^-: E°cell = 1.71 − 1.51 = +0.20 V → spontaneous, so H_2O_2 can oxidise Mn^{2+} to MnO_4^- (in suitable acidic medium).
For PbO_2 vs Mn^{2+}/MnO_4^-: E°cell = 1.455 − 1.51 = −0.055 V → non-spontaneous, so PbO_2 cannot oxidise Mn^{2+} to MnO_4^-.

52) $\wedge_m = \frac{k \times 1000}{M}$
$= \frac{0.0248 \times 1000}{0.2}$
$= 124\ S\ cm^2\ mol^{-1}$

53)

pH = 0
$H^+ = 1$
$E_1 = E^\circ - 0.059 \log\left(\frac{1}{H^+}\right)$
= 0
HCl + NaOH
SA SB
pH = 7
$H^+ = 10^{-7}$
$E_2 = -0.059 \log 10^7$
$= -0.413$
So E decrease by 0.413

54) $(\lambda_M^o)_{Al_2(SO_4)_3} = 2(\lambda_M^o)_{Al^{+3}} + 3(\lambda_M^\infty)_{SO_4^{-2}}$
$858 = 2(\lambda_M^o)_{Al^{+3}} + 3 \times 160$
$(\lambda_M^o)_{Al^{+3}} = 189\ S \cdot cm^2 mol^{-1}$

55) Pressure of air = 750 – 100 = 650 mm of Hg
on compressing Pf = 650 × 3 mm of Hg = 1950 mm of Hg
so PT = (1950 + 100) = 2050 mm of Hg

56) In solution (given)
$X_A P_A^o = X_B P_B^o = (1 - X_A) P_B^o$
$\because P_A^o = 0.6$ and $P_B^o = 0.2$
$\Rightarrow 0.6\ X_A = (1 - X_A) \times 0.2$

$\Rightarrow 0.8\ X_A = 0.2$

or $X_A = \frac{0.2}{0.8} = 0.25$

57) $\Delta T_f = i \times K_f \times m$

$0.46 = 1 \times 3.6 \times \frac{1/M_{solute}}{0.1}$

$\therefore M_{solute} = 78g$

$E.f. = C_{92.3/12}\ H_{7.7/1} = C_1H_1$

$\therefore$ M.F. $= (C_1H_1)_6$

58) **Assertion:** NaCl is a non-volatile solute. On dissolving in H_2O, it lowers vapour pressure, so a higher temperature is needed to reach atmospheric pressure → boiling point increases. **(True)**
Reason: Addition of NaCl **decreases** vapour pressure of water, not increases. **(False)**

59) (1). **Considerable interaction** (solvation/hydration) between the dispersed phase and dispersion medium is the characteristic feature (Lyophilic=solvent-loving) (Correct).
(2). The strong interaction/solvation layer makes them very **stable** and **not easily coagulated** (Correct).
(3). Lyophilic sols are generally proteins/organic molecules whose charge depends on the pH. They are not exclusively negative (Incorrect).
(4). Due to strong interaction with water (dispersion medium), the particles are surrounded by water molecules, a process called **hydration** or solvation (Correct).

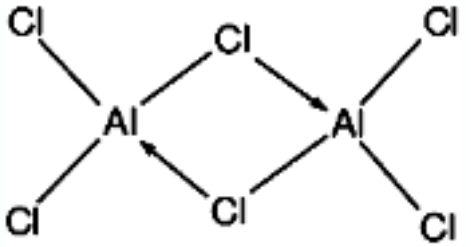
60)

A spontaneous reaction occurs when the free element is a stronger oxidizing agent than the element in the compound.
• From (1) : Y_2 oxidizes X^-, so Y_2 is more reactive than X_2.
• From (2) : Y_2 cannot oxidize W^-, so W_2 is more reactive than Y_2.
• From (3) : X_2 oxidizes Z^-, so X_2 is more reactive than Z_2.
The order of oxidizing power is :
$W_2 > Y_2 > X_2 > Z_2$.
Since the standard oxidation potential (E°) follows the opposite trend of oxidizing power :
$E^o_{W^-/W_2} < E^o_{Y^-/Y_2} < E^o_{X^-/X_2} < E^o_{Z^-/Z_2}$

61) **Explanation -** The question asks you to identify the correct chemical formula for the acetylide ion from a list of options, focusing on its charge and composition.
Concept - (Hydrolysis of Metal Carbide)
The acetylide ion is C_2^{2-}. It's formed when acetylene loses both protons, resulting in a carbon-carbon triple bond with a -2 charge.
Final Answer - Option (3)

62) **Explain Question:** Identify incorrect dimerisation match **Concept:** Dimerisation bonding.
Solution:

A. NO_2 dimerises to N_2O_4 via normal 2C–2e⁻ bond (Correct)
B. BH_3 forms B_2H_6 via 3C–2e⁻ bonds (Correct)
C. $AlCl_3$ forms Al_2Cl_6 via 3C–4e⁻ (coordinate bridges), not 2C–2e⁻ (Incorrect)



A. CH_3COOH dimerises via hydrogen bonding (Correct)

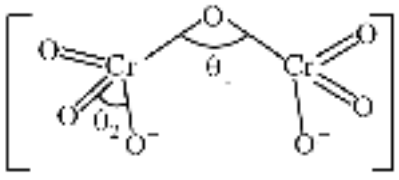
63)

Question Analysis/Explanation:
The question asks which of the provided statements are direct consequences of the inert pair effect, a phenomenon in heavier p-block elements.
Concept: Inert pair effect
Answer Explanation:

(d) CCl_2 is unstable but $PbCl_2$ is stable: This is a direct consequence of the inert pair effect. Carbon (C) doesn't exhibit the inert pair effect, so its +2 oxidation state is unstable. Lead (Pb), being lower in the group, shows the effect, making $PbCl_2$ stable.
(e) PbO_2 is a good oxidizing agent: Due to the inert pair effect, Pb^{+4} is less stable than Pb^{+2}. Therefore, PbO_2 readily gets reduced to Pb^{+2}, acting as a strong oxidizing agent.
(a) $SnCl_4$ act as a reducing agent: $SnCl_4$ is already in its higher oxidation state (+4) and does not have a tendency to lose electrons. Therefore, it does not act as a reducing agent.
(b) $SnCl_2$ act as an oxidizing agent: $SnCl_2$ can act as a reducing agent, not oxidizing, due to the inert pair effect, it tends to go to $SnCl_4$.
(c) SnO_2 is amphoteric: SnO_2 amphoteric nature is not directly related to the inert pair effect.
Therefore, options (d) and (e) are directly related to the inert pair effect.

Final Answer - Option (2)

64)



$\theta_1 > \theta_2$

65)

Enthalpy of atomisation Cr > Zn
Metallic bond strength ∝ number of unpaired electrons

66)

$Ce^{+4} = {}_{54}[Xe]$
$Yb^{+2} = {}_{54}[Xe]\,4f^{14}5d^{0}$
Both species do not contain any unpaired electron. Hence, they are diamagnetic.

67)

Covalent Radius : Mn < Cr ; Density : Cu > Zn
Metallic Radius : Mn > Cr

68)

Question explanation: This question is asking about EAN rule and complex reactivity.
Concept: This question is based on stable complexes have 18-electron (EAN 36) configuration, otherwise reactive.
Solution : $[Fe(CO)_5]$ has EAN = 36, already stable, cannot readily gain or lose electrons. Therefore, it cannot act as oxidising or reducing agent. Other complexes either dimerise or accept/donate electrons to reach stable EAN.
Final answer: Option 3

69)

As per structure
(I) $[Ma_3b_3]$ type
(II) $[M(AA)_3]$

70)

S^{2-} (Anion)

71)

C-C bond length
C_2H_4 < $K[PtCl_3(C_2H_4)]$

72)

S^{2-} (Anion)

73)

Nessler reagent
⇒ K_2HgI_4 + KOH

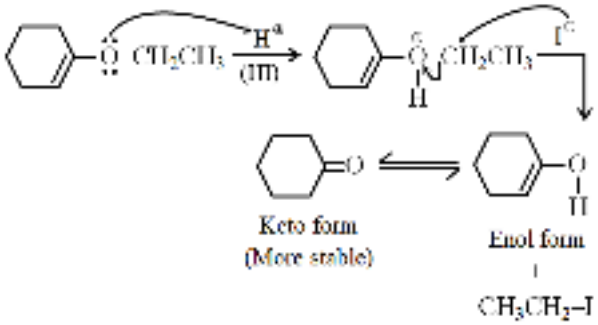
74) NH_4^+ : Gives pungent smell gas with NaOH

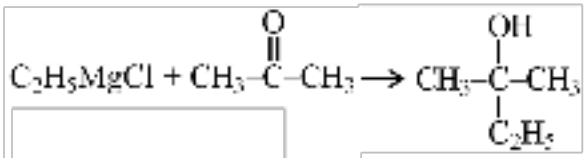
Fe^{+3} : Gives red-brown ppt with NH_4OH
Cd^{+2} : Gives yellow ppt with H_2S
Ag^{+} : Gives white ppt with dil HCl

75)

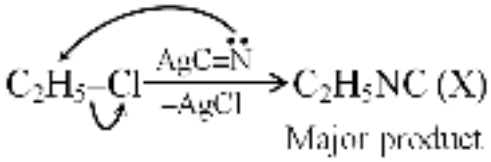
Pb^{+2}, Cu^{+2} : H_2S + HCl
Al^{+3}, Fe^{+3} : NH_4OH + NH_4Cl
Co^{+2}, Ni^{+2} : H_2S + NH_4OH
Ca^{+2}, Ba^{+2} : $(NH_4)_2CO_3$ + NH_4OH

76) Reactant has no tertiary radical hence it follows S_N2 mechanism.



77) 

78) $CH_3CH_2CN \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ SnCl_2\ \ HCl} CH_3CH_2CHO$

79) 

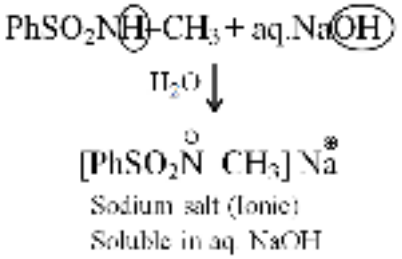
But its functional isomer is C_2H_5CN, hence ans. (2)

80)

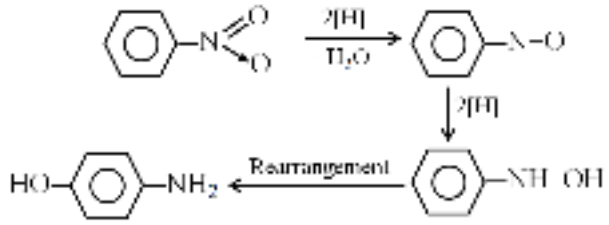
$NaBH_4$ is a mild reducing agent. It does not reduce –COOH. It reduce –CHO and >C=O (ketone) in 1° alcohol and 2° alcohol respectively.

81) Rate of decarboxylation ∝ Stability of carbanion
Apply –I of $–NO_2$ group because it has cyclohexane not benzene.

82) It is based on Hinsberg test. If acidic H-atom is bonded with N atom. Then it forms ionic salt i.e., soluble in aq. NaOH.



83)



84) NaOH H_2SO_4

$N_1V_1 = N_2V_2 \rightarrow$ volume of H_2SO_4 neutralized by NaOH

$$\frac{1}{10} \times 100 = \frac{1}{2} \times V_2$$

$V_2 = 20$ mL

Hence, volume of H_2SO_4 neutralized by $NH_3(V)$ = given volume – V_2

$V = 60 - 20 = 40$ mL

$$\% \text{ of N} = \frac{1.4 \times N \times V}{\text{Mass of org.compd.}}$$

$$= \frac{1.4 \times 1 \times 40}{2 \times 0.5} = 56\%$$

85) Presence of C-atom is compulsory for test of 'N' because 'N' is converted in NaCN in Lassaigne's test.

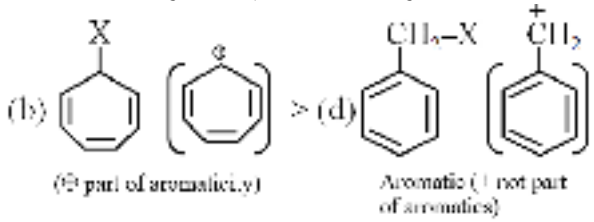
86) Factual

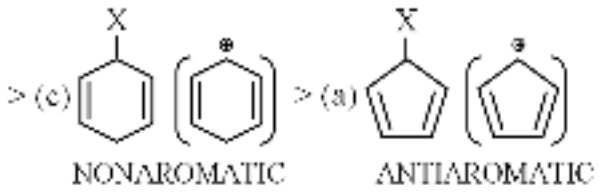
Lactose = β-D-galactose + β-D-glucose

87) Factual

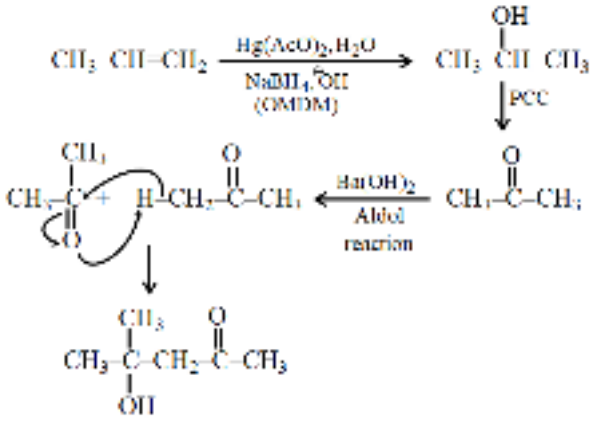
88) Both phenyl acetaldehyde and acetaldehyde give positive test with (2) and (3).

89) Reactivity of $S_N1 \propto$ Stability of $C^{\oplus}$ intermediate





90)



BIOLOGY

91) NCERT Pg. # 74, 75, 76

92) NCERT-XI, Pg. # 73

93) NCERT-XI, Pg. # 73, 74, 75, 76

94) NCERT XI Pg # 60, 61

95)

NCERT-XI, Pg. # 62, 65

96) NCERT-XI, Pg # 60

97) NCERT-XI, Pg. # 69

98) NCERT Pg.# 68

99) NCERT-XII, Pg # 13

100) NCERT-XII, Pg. No. 13

101) NCERT-XII, Pg. # 18

102) NCERT-XII, Pg. # 10

103) NCERT-XII, Pg. # 6

104)

NCERT-XII, Pg # 10

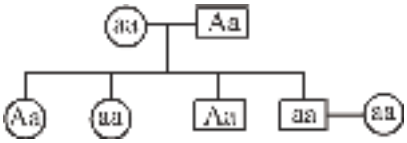
105)

NCERT XII Page # 14, Fig. # 1.11(a)

106)

The chromosomal theory of inheritance states that chromosomes carry genes and their behavior explains mendelian inheritance.
The old statement is -3
because
(1) In diploid cells, both chromosomes and genes occur in pairs.
(2) In triploid cells, there are three sets of chromosomes and genes are in three copies, not just genes alone in pairs.

107) Autosomal dominant :-



108) **Question Explanation:** The question asks that which of the statements is wrong/ Incorrect.
Concept: Mendelism**.**
Solution:
The incorrect statement is:
“A heterozygous organism always has a different phenotype than a homozygous organism.”
This is incorrect because in cases of complete dominance, a heterozygous (Aa) organism has the same phenotype as a homozygous dominant (AA) organism.
• Example: In Mendelian inheritance, Aa (heterozygous) and AA (homozygous dominant) both show the dominant phenotype.
• This statement would be true only for incomplete dominance or codominance, where heterozygous individuals show a different phenotype.
So, the correct answer is: 3.
Final answer: 3.

109) NCERT-XII, Pg. # 74

110)

Explain Question:
The question asks about the percentage of pure plants in F2 generation of Mendel's monohybrid cross.
Concept:
Monohybrid Cross.
Solution:
The percentage of pure plants in F2 generation of Mendel's monohybrid cross. [TT and tt are pure (homozygous)] → 50 %
So, the correct answer is :2
Final answer: Option (2)

111)

NCERT Pg. # 54, 56

112) NCERT-XII, Pg. # 90

113) NCERT-XII, Page # 91

114)

NCERT Pg. No. 83 & 84

115) **Question Explanation:** Identifying incorrect statements about VNTRs.
Concept: VNTR (Variable Number of Tandem Repeats)
Solution:(i) Incorrect; VNTRs are a class of minisatellite DNA, not microsatellite(iv) Incorrect; the number of repeats in a VNTR shows a very high degree of polymorphism, which is why it is useful for DNA fingerprinting. Statements (ii) and (iii) are correct.
Final Answer: 4. (i) and (iv)

116) **Question Explanation:** Ratio of DNA types after two generations of replication.

Concept: Meselson and Stahl Experiment

Solution:

- Generation 0: 15N15N (Heavy)

- Generation 1: 15N14N (Intermediate)

- Generation 2: Replication of 15N14N produces one 15N14N and one 14N14N per molecule.

- Ratio (N14N15 : N14N14 : N15N15): 1 : 1 : 0

Final Answer : Option **(1)**

117) NCERT-XII, Pg. # 103

118) NCERT-XII, Pg No. # 91, 92, 93, 94, 95

119) NCERT Pg # 96

120)

Question Explanation: Predicting the result of gel electrophoresis.
Concept: Gel Electrophoresis

Solution: Gel electrophoresis separates DNA fragments based on size; smaller fragments migrate faster and further towards the positive electrode. From the diagram of the original DNA, fragment 'b' is the smallest, 'a' is medium, and 'c' is the largest. Therefore, on the gel, 'c' will be closest to the start (negative electrode), 'a' will be in the middle, and 'b' will be the farthest. This pattern is shown in option 2.
Final Answer: 2.

121)

NCERT-XII, Pg. # 213

122)

NCERT-XII, Pg. # 206

123)

NCERT-XII, Pg. # 207

124)

NCERT-XI, Pg. # 210

125)

NCERT-XII, Pg. # 210

126)

NCERT-XII, Pg. # 200, 201

127) NCERT-XII Pg. # 200

128)

NCERT-XII, Pg. # 197

129)

NCERT-XII, Pg # 197-200

130) NCERT-XII, Pg.# 197

131) NCERT-XII, Pg. # 222

132) NCERT, Pg. # 219

133) NCERT-XII, Pg. # 222, 223

134)

NCERT-XII, Pg. # 223

135) NCERT-XII, Pg. # 225

136) NCERT XII Pg # 222

137)

NCERT Pg. # 168

138) NCERT, Pg. # 191

139)

NCERT Pg. # 170, 171

140) NCERT Pg # 203 (E), 189 (H)

141) NCERT Pg # 200,210,211

142)

NCERT XII, Pg. # 182

143) NCERT Pg. # 177

144) NCERT Pg. # 182-183

145) NCERT, Pg. # 179

146) NCERT - 208 (12.1)

147) NCERT Pg. # 182

148) XII NCERT Pg # 187

149) NCERT XII, Pg. # 181

150)

NCERT, Pg. # 150,153

151)

NCERT-XII, Pg. # 124, 125

152) NCERT XII (English)
Page-115, 117

153) NCERT Pg. # 120, 121

154)

NCERT Pg. # 113

155) NCERT, Pg. # 118

156)

NCERT Pg. # 113

157)

NCERT Pg No. 117

158) NCERT Pg # 119

159) NCERT Pg. no. # 34

160) NCERT, Pg # 36

161) NCERT XII Pg # 26,27,28,29

162) NCERT Pg#35,38(E)

163) NCERT P.No. 35

164) NCERT Pg. # 37

165) NCERT Pg. # 32

166) NCERT Pg. # 37

167)

NCERT-XII, Pg. # 54

168) NCERT Pg. # 45, 46

169) NCERT-XII, Pg. # 23

170) NCERT Pg. # 46

171)

NCERT Pg. # 47

172) NCERT_Pg. # 130-131

173)

NCERT Pg. # 131, 132

174) NCERT Pg. # 134-135

175) NCERT Pg. # 141

176) NCERT Pg. # 135

177) NCERT_Pg. # 135-174

178) NCERT Pg. # 137

179)

NCERT Pg. # 143

180)

NCERT Pg. # 136